

Twisted Stieltjes Convolutions and Harmonic Laurent Identities

Periodic Stieltjes constants, polygamma contact terms,
polylogarithm order derivatives, and shifted harmonic sums

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Abstract

We continue the exact-identity programme of the ProveIt polylogarithm manuscript in two directions. First, we answer an archived research question about the all-index half-twist Stieltjes multiplication law, including point-supported counterterms. A meromorphic periodic Lerch family gives the construction for every real nonintegral twist. Its convolution identity is the full Dirac mass, and its covariant derivative correction is an explicit harmonic-number multiple of a Dirac derivative. Rational twists reduce to finite Hurwitz and Stieltjes combinations. At the half-twist, an ordinary convergent subtracted digamma integral evaluates in terms of $\log \Gamma(1/4)$, Euler's constant, and logarithms. An exact zero-frequency subtraction controls the untwisted limit.

Second, a two-variable Mellin continuation determines every principal Laurent coefficient and finite part of $\sum_{n \geq 1} H_{n-1}(\{1\}^r)/(n+a)^s$ at $s = 1$ and at all nonpositive integers. Gamma and generalized Bernoulli coefficients give a finite formula at every harmonic depth and complex outer shift. We derive residue-aware differentiation, depth-mixing reflection, rational multiplication, and complete antiderivative ladders for these coefficient polynomials, with explicit half-shift examples.

For self-contained proofs and consistent normalizations we consolidate the untwisted periodic Stieltjes algebra and polylogarithm kernels already developed in archived repository continuations, crediting those results and their classical antecedents. The package includes exact symbolic checks, independent numerical diagnostics, a source audit, two proposed manuscript corrections, and further research questions. No universal priority claim is made; the remaining S_6 and newer S_8 special-value conjectures remain unresolved.

Source baseline: e0d9463bdee9685dfb1dddb819059cc738540c57.

The archive contains this article, its complete $\text{\LaTeX}$ source, reproducible verification code, a source manifest, and proposed integration patches.

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1 Scope, conventions, and the status of the results

This continuation develops two exact extensions of the ProveIt programme. First, it answers the archived question about an all-index half-twist Stieltjes multiplication law with its endpoint terms. The twist changes the convolution identity from $\delta_0 - 1$ to δ_0 . Second, it determines the principal Laurent coefficients and finite parts of a shifted harmonic Dirichlet family at every nonpositive integer, with reflection, rational multiplication, and antiderivative formulas for those coefficients.

Both developments require a precise normalization at a moving pole. The periodic correction is supported at the endpoint of a circle. The harmonic correction is a Laurent residue, which survives when finite parts are differentiated. Their common role explains why a formula at regular parameters cannot simply be specialized or differentiated at a singular parameter.

1.1 The inspected baseline

We froze the repository at e0d9463bdee9685dfb1dddb819059cc738540c57, whose commit is dated 10 October 2026, 21:44:12 UTC. The audit used the complete set of 72 chapter sources and the eight non-PDF manuscript support files. The pending archive ProveIt_Polylogarithm_Stieltjes_Continuation_2026-10-10.zip was matched to its Git blob and its mathematical source was inspected. Five relevant archived report sources were also checked. The machine-readable source manifest records exact paths and hashes. These statements describe the inspected set; they do not assert that every report or historical archive in the repository was reviewed. Manuscript and archival status are kept separate [1, 2].

The archived *Convolution Algebra* report explicitly asks for a nonintegral-twist normal form and, as a first target, the all-index half-twist multiplication table with point-supported counterterms [3, final research section]. Section 4 supplies this normalization, multiplication law, and covariant derivative/primitive calculus. It gives a rational-twist conversion to Hurwitz and Stieltjes coordinates, a convergent half-twist integral, and the exact zero-frequency correction needed in the limit as the twist tends to zero.

The same archive and two further continuations already contain the untwisted periodic Bell algebra, polygamma contact terms, normalized primitive tower, and reflected polylogarithm kernels [3, 4, 5]. Those results are consolidated with independent complete proofs here; they are not new theorems relative to the pinned repository. The incoming Stieltjes continuation supplies additional ordinary translation and integration formulas [2]. Earlier alternating-harmonic reports likewise already use beta representations, cyclotomic projections, and scalar Mellin subtraction [6, 7].

The source's S_2 and S_4 identities already have exact proofs. The frozen S_6 target and the newer S_8 candidate remain conjectural. Our proofs do not evaluate either of them. The half-twist target answered here is an explicit research question; it is distinct from those two special-value conjectures.

1.2 Classical inputs and the contribution of this article

The Fourier coefficients of Hurwitz zeta, Gamma-beta identities, and

$$\zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$$

are classical [8, 9]. Normalized Hurwitz convolution is established; Sun states an ordinary-integral form in his study of invariant functions [13, Remark 3.2 in the arXiv version]. Periodic complex-power distributions provide the natural background [14]. Kim’s Hurwitz-transform continuation is a recent antecedent for scalar test-function subtraction [15].

Young’s parametric Euler sums supply the beta-derivative evaluation framework [18]. His height-one work and unshifted Laurent study are directly relevant [17, 16]; the harmonic Stieltjes literature concerns related, but not identically shifted, families [19]. The full text of Young’s 2023 paper was unavailable. Its inspected publisher abstract already rules out a broad priority claim for unshifted Laurent expansions.

The additions for this continuation are the fully normalized twisted extension and the joint coefficient theorem for the outer-shifted harmonic family, with its finite polynomial calculus. Beta differentiation, Mellin continuation, and the unshifted height-one theory are treated as antecedents. Some consequences may have equivalent formulations elsewhere. A result described as *proved here* has a complete analytic proof in this article; a result described as an *addition to the inspected source* was not located in that source set. Neither phrase asserts universal literature priority. Proof and provenance statuses are recorded separately.

Result or ingredient	Status and use
Untwisted periodic calculus	Existing repository foundation, independently checked; all endpoint terms and integration constants retained.
Twisted Stieltjes calculus	Answers the archived half-twist target, for every real nonintegral twist, with the singular normalization included.
Ordinary half-twist integral	A convergent consequence with a midpoint evaluation in $\log \Gamma(1/4)$, Euler’s constant, and logarithms.
Harmonic beta evaluations	Established antecedent connecting harmonic polynomials, Gamma derivatives, and positive integer values.
Shifted harmonic Laurent data	Complete principal and finite parts; residue-aware differentiation, reflection, rational multiplication, and repeated primitives.
Numerical checks	Independent diagnostics; none replaces an analytic proof.

1.3 Notation and normalizations

Order derivatives always concern the first argument:

$$\zeta^{[j]}(s, a) = \partial_s^j \zeta(s, a), \quad \text{Li}_s^{[j]}(z) = \partial_s^j \text{Li}_s(z).$$

By contrast, $\psi^{(k)}(a)$ differentiates the argument a , and

$$\zeta(1 + w, a) = \frac{1}{w} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) w^n. \quad (1.1)$$

Thus $\gamma_0(a) = -\psi(a)$, and $\gamma = \gamma_0(1)$. Finite parts are Laurent constant terms, unless a displayed cutoff definition specifies the equivalent integral normalization.

On $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, write

$$\hat{T}(n) = \langle T, e^{-2\pi i n x} \rangle, \quad \Delta = \delta_0 - 1.$$

The distribution Δ is the identity for convolution on the subspace of distributions with zero mean. The full algebra has identity δ_0 . Reflection means $\check{T}(x) = T(-x)$, and D denotes the distributional

argument derivative. A convolution is always the compact-circle convolution. It is defined for any pair of distributions on the circle; it must not be confused with a pointwise product of singular distributions. All logarithms of $2\pi in$, for $n \neq 0$, mean

$$\log(2\pi in) = \log(2\pi|n|) + \frac{\pi i}{2} \operatorname{sgn}(n).$$

This branch convention is part of the identities.

For finite harmonic sums we use

$$H_N^{(j)} = \sum_{k=1}^N k^{-j}, \quad H_N(\{1\}^r) = \sum_{N \geq n_1 > \dots > n_r \geq 1} \frac{1}{n_1 \cdots n_r}, \quad H_N(\{1\}^0) = 1.$$

Their generating polynomial is

$$\sum_{r \geq 0} H_N(\{1\}^r) u^r = \prod_{k=1}^N (1 + u/k).$$

Thus all depths concern explicit polynomials in generalized harmonic numbers:

$$H_N(\{1\}^2) = \frac{H_N^2 - H_N^{(2)}}{2}, \quad H_N(\{1\}^3) = \frac{H_N^3 - 3H_N H_N^{(2)} + 2H_N^{(3)}}{6}.$$

2 Periodic finite parts and harmonic contact terms

We begin by consolidating the untwisted foundation already present in three archived ProveIt continuations [3, 4, 5]. The endpoint normalization is fixed by analytic continuation from the integrable Hurwitz family. Its spectral finite parts and argument derivatives differ by an explicit harmonic-number multiple of a Dirac mass derivative. This distinction will persist in the twisted extension.

The normalized Hurwitz convolution is classical [13, Remark 3.2]; complex-power distributions provide related background [14, Section 5]. Scalar meromorphic continuation after pairing with a smooth function is also covered by Kim's Hurwitz-transform theorem [15, Theorem 1 and Corollary 2]. The full proofs below establish the conventions used in the rest of this article and independently check the archived formulas; the untwisted Bell algebra and primitive tower are not new-to-repository claims.

2.1 Conventions and the meromorphic periodic family

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, use normalized Lebesgue measure, and put

$$\hat{U}(n) = \langle U, e^{-2\pi i n x} \rangle, \quad D = \frac{d}{dx}, \quad \Delta = \delta_0 - 1.$$

The Dirac mass has Fourier coefficient 1 at every integer. Thus $\hat{\Delta}(0) = 0$ and $\hat{\Delta}(n) = 1$ for $n \neq 0$. On the convolution algebra of distributions with zero mean, Δ is the identity. Reflection is $\check{U}(x) = U(-x)$, so $\hat{\check{U}}(n) = \hat{U}(-n)$. Throughout,

$$L_n = \log(2\pi|n|) + \frac{\pi i}{2} \operatorname{sgn} n, \quad n \neq 0. \quad (2.1)$$

Consequently $(2\pi in)^z$ always means $\exp(zL_n)$.

Theorem 2.1 (The entire normalized family and all residues). *There is an entire distribution-valued function W_s characterized by*

$$\widehat{W}_s(0) = 0, \quad \widehat{W}_s(n) = (2\pi in)^{s-1} \quad (n \neq 0). \quad (2.2)$$

The family

$$Z_s = \Gamma(1-s)W_s \quad (2.3)$$

is the unique meromorphic continuation, as a periodic distribution, of $x \mapsto \zeta(s, x)$ from $\operatorname{Re} s < 1$. Its poles are simple and occur at every positive integer. Their residues are

$$\operatorname{Res}_{s=1} Z_s = 1 - \delta_0 = -\Delta, \quad (2.4)$$

$$\operatorname{Res}_{s=k+1} Z_s = \frac{(-1)^{k+1}}{k!} \delta_0^{(k)} \quad (k \geq 1). \quad (2.5)$$

For all complex s, t , the identities

$$DW_s = W_{s+1}, \quad W_s * W_t = W_{s+t-1} \quad (2.6)$$

hold. Equivalently, as meromorphic distribution identities,

$$DZ_s = -sZ_{s+1}, \quad Z_s * Z_t = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} Z_{s+t-1}. \quad (2.7)$$

At intersections of polar divisors the second expression is interpreted by its equal Gamma-normalized expression, before coefficients are taken.

Proof. On every compact set of s the coefficients in (2.2) have uniform polynomial growth in $|n|$. Every fixed order derivative in s adds only a power of L_n . Since the Fourier coefficients of a smooth test function decrease faster than every inverse power, the series

$$\langle W_s, \phi \rangle = \sum_{n \neq 0} (2\pi in)^{s-1} \widehat{\phi}(-n)$$

and all its fixed order derivatives converge normally on compact sets. This constructs the entire family. Hurwitz's classical Fourier formula [8, §25.11] identifies (2.3) with $\zeta(s, x)$ for $\operatorname{Re} s < 0$. The relation $\zeta(s, x) = x^{-s} + \zeta(s, 1+x)$ shows that the latter is holomorphic with values in $L^1(0, 1)$ on $\operatorname{Re} s < 1$, so the identification extends throughout that half-plane. The identity theorem for each test-function pairing gives uniqueness of the continuation.

At $s = k + 1 + w$,

$$\Gamma(-k-w) = \frac{(-1)^{k+1}}{k!w} + \frac{(-1)^k}{k!} (H_k - \gamma) + O(w), \quad H_0 = 0. \quad (2.8)$$

The Fourier coefficients show that $W_1 = \Delta$ and $W_{k+1} = D^k \Delta = \delta_0^{(k)}$ for $k \geq 1$. This proves the residues and also shows that none of the listed poles is removable. Multiplication of Fourier coefficients proves (2.6). Convolution of two distributions on the compact group $\mathbb{T}$ is well defined. Normal convergence after pairing with a test function justifies the parameter identities jointly. Finally, $\Gamma(1-s) = -s\Gamma(-s)$ proves the derivative identity in (2.7). $\square$

The residue at $s = 1$ has both a constant and a Dirac term. The constant is the ordinary Hurwitz pole seen on every open subinterval of $(0, 1)$; the Dirac term comes from the nonintegrable endpoint x^{-s} . Omitting either term destroys the zero Fourier coefficient. The poles at $2, 3, \dots$ are entirely supported at the endpoint and are invisible in the pointwise meromorphic function of s for a fixed $0 < x < 1$.

2.2 Cutoff definitions and the normalization at the endpoint

Define T_n by the Laurent expansion

$$Z_{1+w} = -\frac{\Delta}{w} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} T_n w^n. \quad (2.9)$$

Each T_n agrees with $\gamma_n(x)$ on $0 < x < 1$. Define also

$$F_k = \text{FP}_{s=k+1} Z_s, \quad P_k = (-1)^{k+1} k! F_k \quad (k \geq 0), \quad (2.10)$$

where FP denotes the constant Laurent coefficient. Thus $F_0 = T_0$ and $P_0 = -T_0$. For $k \geq 1$, the usual $\psi^{(k)}(x) = (-1)^{k+1} k! \zeta(k+1, x)$ shows that P_k agrees with the ordinary k th polygamma function away from 0.

Proposition 2.2 (Equivalent Hadamard finite parts). *For every smooth periodic test function ϕ ,*

$$\langle T_n, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_{\epsilon}^1 \gamma_n(x) \phi(x) dx + \frac{\phi(0)}{n+1} \log^{n+1} \epsilon \right\}. \quad (2.11)$$

For $k \geq 1$,

$$\langle F_k, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_{\epsilon}^1 \zeta(k+1, x) \phi(x) dx - \sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} \frac{\epsilon^{j-k}}{k-j} + \frac{\phi^{(k)}(0)}{k!} \log \epsilon \right\}. \quad (2.12)$$

All these distributions have zero mean.

Proof. Subtract a Taylor polynomial of the test function at 0. For any $N \geq 0$, the pairing of Z_s has the continuation

$$\begin{aligned} \langle Z_s, \phi \rangle &= \int_0^1 x^{-s} \left(\phi(x) - \sum_{j=0}^N \frac{\phi^{(j)}(0)}{j!} x^j \right) dx + \sum_{j=0}^N \frac{\phi^{(j)}(0)}{j!(j+1-s)} \\ &\quad + \int_0^1 \zeta(s, 1+x) \phi(x) dx, \end{aligned} \quad (2.13)$$

in $\text{Re } s < N+2$, apart from the displayed and Hurwitz poles. Taking the constant coefficient at $s = k+1$, with $N = k$, proves (2.12) by elementary integration of the Taylor monomials. At $s = 1+w$, use

$$\gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1+x).$$

The coefficient in (2.13) is

$$\int_0^1 \frac{\log^n x}{x} (\phi(x) - \phi(0)) dx + \int_0^1 \gamma_n(1+x) \phi(x) dx.$$

This is precisely (2.11). The zero-mean assertion follows from (2.2) by taking Laurent coefficients. $\square$

Zero mean alone does not specify a periodic extension of a singular function: derivatives of a Dirac mass have zero mean as well. Here the normalization is fixed by (2.3), or equivalently by the cutoff formulas. This qualification is essential in using the word “canonical.”

2.3 The harmonic contact correction

Theorem 2.3 (Spectral finite parts and compatible derivatives). *For $n \neq 0$ and $k \geq 0$,*

$$\widehat{P}_k(n) = (2\pi in)^k (\gamma + L_n - H_k), \quad \widehat{P}_k(0) = 0. \quad (2.14)$$

Consequently,

$$DP_k = P_{k+1} + \frac{1}{k+1} \delta_0^{(k+1)}. \quad (2.15)$$

If

$$g(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi), \quad 0 < x < 1,$$

is periodized as a locally integrable function, then $Dg = P_0$. The derivative-compatible extensions

$$Q_k = D^{k+1}g = D^k P_0 \quad (2.16)$$

satisfy $DQ_k = Q_{k+1}$ exactly, and

$$Q_k = P_k + H_k \delta_0^{(k)} \quad (k \geq 1), \quad Q_0 = P_0. \quad (2.17)$$

Proof. Expand $W_{k+1+w} = W_{k+1} + w \partial_s W_s|_{s=k+1} + O(w^2)$ against (2.8); after multiplication by $(-1)^{k+1}k!$, the finite Fourier coefficient is $(2\pi in)^k (\gamma + L_n - H_k)$. Since $H_{k+1} - H_k = 1/(k+1)$, differentiation of this coefficient gives (2.15). Lerch's identity gives $g = \partial_s Z_s|_{s=0}$. Taking the first coefficient in $DZ_s = -sZ_{s+1}$ at $s = 0$ gives $Dg = -T_0 = P_0$; the coefficient at degree zero separately says $DZ_0 = \Delta$. Finally the Fourier coefficient of $D^k P_0 - P_k$ is $H_k (2\pi in)^k$, which is (2.17). $\square$

For example, the extension given by the finite part of $\zeta(s, x)$ at $s = 2$ yields a periodic trigamma P_1 , whereas $D^2 g = P_1 + \delta'_0$. Both agree with $\psi'(x)$ on $(0, 1)$. The discrepancy is supported precisely at the identified endpoints; it is not a failure of ordinary differentiation on the open interval.

3 Stieltjes convolution identities and repeated primitives

3.1 All Stieltjes functions as convolution polynomials

The regularized germ

$$B(u) = -uZ_{1+u} = \Gamma(1-u)W_{1+u}, \quad |u| < 1, \quad (3.1)$$

satisfies

$$B(u) = \Delta + \sum_{r \geq 1} \frac{(-1)^r}{(r-1)!} T_{r-1} u^r. \quad (3.2)$$

The following formula is a statement in the commutative convolution algebra with identity Δ ; exponential and Bell-polynomial expressions can equivalently be read as formal power series.

Theorem 3.1 (Stieltjes convolution generator). *For $|u| < 1$,*

$$B(u) = \exp_* \left(uP_0 + \sum_{r \geq 2} \frac{\zeta(r)}{r} u^r \Delta \right). \quad (3.3)$$

Here the series defines the analytic germ after pairing with any smooth test function. In particular, if $\mathcal{B}_j^*$ denotes the complete exponential Bell polynomial with every multiplication interpreted as convolution and every scalar constant as that scalar times Δ , then

$$T_n = \frac{(-1)^{n+1}}{n+1} \mathcal{B}_{n+1}^*(P_0, \zeta(2)\Delta, 2!\zeta(3)\Delta, \dots, n!\zeta(n+1)\Delta). \quad (3.4)$$

The two-parameter form is

$$B(u) * B(v) = C(u, v)B(u+v), \quad C(u, v) = \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)}, \quad (3.5)$$

as an analytic germ at $(u, v) = (0, 0)$. Its scalar correction has

$$\log C(u, v) = \sum_{r \geq 2} \frac{\zeta(r)}{r} (u^r + v^r - (u+v)^r). \quad (3.6)$$

Thus every $T_m * T_n$ is a finite explicit linear combination of $\Delta, T_0, \dots, T_{m+n+1}$, with coefficients polynomial in positive integer zeta values.

Proof. At a nonzero Fourier mode, $\widehat{P}_0(n) = \gamma + L_n$. The Taylor expansion

$$\log \Gamma(1-u) = \gamma u + \sum_{r \geq 2} \frac{\zeta(r)}{r} u^r$$

therefore proves (3.3) mode by mode. The same polynomial growth argument as in Theorem 2.1 proves normal convergence in every smaller disk after test-function pairing. Comparing the complete Bell-polynomial generating function with (3.2) gives (3.4); the factorial is $n!/(n+1)! = 1/(n+1)$. Equation (3.5) follows from (2.6). The logarithm of its Gamma quotient is (3.6); its linear Euler-constant terms cancel. Coefficient extraction at $u^{m+1}v^{n+1}$ gives the last assertion. $\square$

The first identities are

$$T_0 = -P_0, \quad (3.7)$$

$$T_1 = \frac{1}{2}(P_0^{*2} + \zeta(2)\Delta),$$

$$T_2 = -\frac{1}{3}(P_0^{*3} + 3\zeta(2)P_0 + 2\zeta(3)\Delta), \quad (3.8)$$

$$T_3 = \frac{1}{4}(P_0^{*4} + 6\zeta(2)P_0^{*2} + 8\zeta(3)P_0 + (3\zeta(2)^2 + 6\zeta(4))\Delta). \quad (3.9)$$

Here P_0^{*r} is an r -fold convolution, not a pointwise power. Eliminating these powers gives, for example,

$$T_0 * T_0 = 2T_1 - \zeta(2)\Delta, \quad (3.10)$$

$$T_0 * T_1 = \frac{3}{2}T_2 - \zeta(2)T_0 + \zeta(3)\Delta, \quad (3.11)$$

$$T_0 * T_2 = \frac{4}{3}T_3 - 2\zeta(2)T_1 + 2\zeta(3)T_0 - 2\zeta(4)\Delta, \quad (3.12)$$

$$T_1 * T_1 = T_3 - 2\zeta(2)T_1 + 2\zeta(3)T_0 - \frac{1}{4}\zeta(4)\Delta. \quad (3.13)$$

The last coefficient uses $\zeta(2)^2 = \frac{5}{2}\zeta(4)$.

3.2 The reflected Stieltjes generator

Define the periodic Hilbert transform on zero-mean distributions by

$$\widehat{\mathcal{H}\hat{U}}(n) = -i \operatorname{sgn}(n) \hat{U}(n) \quad (n \neq 0), \quad \widehat{\mathcal{H}\hat{U}}(0) = 0. \quad (3.14)$$

Its multiplier is bounded, so it is a continuous operator on periodic distributions and commutes with the order coefficient operations above. Reflection in the following theorem acts on the periodic variable only; it does not conjugate the complex parameters.

Theorem 3.2 (All-order reflected convolution). *As an analytic germ at $(u, v) = (0, 0)$,*

$$B(u) * \check{B}(v) = C(u, v) (\cos(\pi v) \operatorname{Id} + \sin(\pi v) \mathcal{H}) B(u + v), \quad (3.15)$$

with the same Gamma quotient $C(u, v)$ as in (3.5). In particular,

$$T_0 * \check{T}_1 = \frac{1}{2} T_2 + \check{T}_2 + \zeta(2)(T_0 + \check{T}_0) + \zeta(3)\Delta, \quad (3.16)$$

$$T_1 * \check{T}_1 = \frac{1}{2}(T_3 + \check{T}_3) + 2\zeta(2)(T_1 + \check{T}_1) + \zeta(3)(T_0 + \check{T}_0) + \frac{7}{2}\zeta(4)\Delta. \quad (3.17)$$

More generally, every $T_m * \check{T}_n$ reduces to a finite linear combination of Δ , T_j and $\check{T}_j$ with $0 \leq j \leq m + n + 1$ and coefficients polynomial in positive integer zeta values over $\mathbb{Q}$.

Proof. Since $L_{-n} = L_n - i\pi \operatorname{sgn} n$, the nonzero Fourier coefficient of the left side of (3.15) is

$$\Gamma(1-u)\Gamma(1-v)e^{(u+v)L_n}e^{-i\pi v \operatorname{sgn} n}.$$

The last factor equals $\cos(\pi v) - i \operatorname{sgn}(n) \sin(\pi v)$, precisely the multiplier of the operator on the right. The normal-convergence argument for the normalized family proves the identity as a distribution-valued analytic germ and justifies every fixed mixed coefficient extraction. This also checks the sign of the Hilbert-transform term directly.

For explicit coefficients put $b = \gamma + \log(2\pi|n|)$ and $a = \frac{\pi i}{2} \operatorname{sgn} n$, so $a^2 = -\pi^2/4$. Equations (3.7)–(3.9) give polynomials $t_j(y) = \hat{T}_j(n)$ in $y = b + a$, while reflection replaces a by $-a$. Direct polynomial multiplication gives

$$\begin{aligned} t_0(b+a)t_1(b-a) &= \frac{1}{2}t_2(b+a) + t_2(b-a) + \zeta(2)(t_0(b+a) + t_0(b-a)) + \zeta(3), \\ t_1(b+a)t_1(b-a) &= \frac{1}{2}(t_3(b+a) + t_3(b-a)) + 2\zeta(2)(t_1(b+a) + t_1(b-a)) \\ &\quad + \zeta(3)(t_0(b+a) + t_0(b-a)) + \frac{7}{2}\zeta(4). \end{aligned}$$

Here $\pi^2 = 6\zeta(2)$ and $\pi^4 = 90\zeta(4)$ have been used. These prove the two displayed identities at every nonzero Fourier mode; all zero modes vanish.

Finally t_j has degree $j + 1$ and nonzero rational leading coefficient $(-1)^{j+1}/(j + 1)$. Put $M = m + n + 2$. After reducing $a^2 = -\pi^2/4$, the product $t_m(b+a)t_n(b-a)$ has the form $A(b) + aE(b)$ with $\deg A \leq M$ and $\deg E \leq M - 1$. The symmetric polynomials $t_j(b+a) + t_j(b-a)$, $0 \leq j < M$, have successive degrees $1, \dots, M$ in b , while their antisymmetric differences divided by a have successive degrees $0, \dots, M - 1$. Together with the constant they form a triangular basis of this space. Elimination divides only by nonzero rational leading coefficients. Its coefficients belong to $\mathbb{Q}[\pi^2, \zeta(2), \dots, \zeta(M)]$, which is the asserted zeta algebra because $\pi^2 = 6\zeta(2)$. $\square$

The frequency-sign contribution can equivalently be encoded by

$$P_0 - \check{P}_0 = -\pi \mathcal{H}\Delta, \quad (P_0 - \check{P}_0)^{*2} = -\pi^2 \Delta.$$

Thus reflection adds a two-dimensional sign algebra to the unreflected Bell calculus, with its constant term fixed by the same endpoint mass.

3.3 Polygamma correlations with all contact terms

Theorem 3.3 (Convolution and reflected convolution). *Let $k, l \geq 0$ and $r = k + l$. With the finite-part normalization (2.10),*

$$P_k * P_l = D^r \{2T_1 + (H_k + H_l)T_0 + (H_k H_l - \zeta(2))\Delta\}, \quad (3.18)$$

$$P_k * \check{P}_l = (-1)^l D^r \{T_1 + \check{T}_1 + H_l T_0 + H_k \check{T}_0 + (H_k H_l + 2\zeta(2))\Delta\}. \quad (3.19)$$

For the derivative-compatible family the harmonic coefficients cancel:

$$Q_k * Q_l = D^r \{2T_1 - \zeta(2)\Delta\}, \quad (3.20)$$

$$Q_k * \check{Q}_l = (-1)^l D^r \{T_1 + \check{T}_1 + 2\zeta(2)\Delta\}. \quad (3.21)$$

Proof. Put $y_n = \gamma + L_n$. The first formula follows from

$$\hat{T}_0(n) = -y_n, \quad 2\hat{T}_1(n) = y_n^2 + \zeta(2)$$

and $(y_n - H_k)(y_n - H_l) = y_n^2 - (H_k + H_l)y_n + H_k H_l$. For reflection set $b_n = \gamma + \log(2\pi|n|)$ and $a_n = \frac{\pi}{2} \operatorname{sgn} n$, so $y_n = b_n + ia_n$ and $y_{-n} = b_n - ia_n$. Then

$$(y_n - H_k)(y_{-n} - H_l) = b_n^2 + a_n^2 - (H_k + H_l)b_n + H_k H_l + ia_n(H_k - H_l).$$

Also

$$\hat{T}_1(n) + \hat{T}_1(-n) = b_n^2 - a_n^2 + \zeta(2),$$

and $H_l \hat{T}_0(n) + H_k \hat{T}_0(-n) = -(H_k + H_l)b_n + ia_n(H_k - H_l)$. The missing constant is $2a_n^2 - \zeta(2) = 2\zeta(2)$. Finally $(2\pi i(-n))^l = (-1)^l (2\pi i n)^l$ accounts for the prefactor in (3.19). The last two identities follow from $Q_k = D^k P_0$ and $D^l \check{P}_0 = (-1)^l D^l \check{P}_0$. $\square$

At the first level,

$$P_0 * \check{P}_0 = T_1 + \check{T}_1 + \frac{\pi^2}{3}(\delta_0 - 1). \quad (3.22)$$

For $0 < a < 1$ its ordinary representative is $\gamma_1(a) + \gamma_1(1-a) - \pi^2/3$. Equation (3.22) supplies the missing endpoint mass in the formal twice-differentiated log-Gamma correlation. In particular, if $C_g = g * \check{g}$, then

$$-D^2 C_g = T_1 + \check{T}_1 + \frac{\pi^2}{3}(\delta_0 - 1). \quad (3.23)$$

The negative sign follows from differentiating the reflected factor.

For $r \geq 1$, the representative of (3.18) on $(0, 1)$ can also be written with one positive-order Hurwitz jet:

$$(P_k * P_l)(a) = (-1)^{r+1} r! \{2\zeta'(r+1, a) + (2H_r - H_k - H_l)\zeta(r+1, a)\}. \quad (3.24)$$

Indeed, differentiating the Hurwitz shift identity at $s = 1$ gives $\gamma_1^{(r)}(a) = (-1)^{r+1} r! [\zeta'(r+1, a) + H_r \zeta(r+1, a)]$. For $r = 0$, the separate constant $\zeta(2)$ from $-\zeta(2)\Delta$ must be retained.

Translations cause no further ambiguity. If $\tau_a U(x) = U(x - a)$ for real a modulo 1, then $(\tau_a U) * (\tau_b V) = \tau_{a+b}(U * V)$ and $\tau_a U * \tau_b \check{V} = \tau_{a-b}(U * \check{V})$. All contact terms above are therefore translated to the sum or difference of the two shifts. General complex translations are not operators on all periodic distributions, because their Fourier multipliers grow exponentially in one direction; they require a separately specified analytic or one-sided boundary-value setting.

3.4 Convergent integral identities without distribution notation

The distribution formulas have ordinary, absolutely convergent integral versions at every nonintegral shift. The following construction makes the subtraction completely explicit. For $m, n \geq 0$ and $0 < a < 1$, put

$$\begin{aligned} \mathcal{I}_{m,n}(a) = & \int_0^a \left\{ \gamma_m(x) \gamma_n(a-x) - \gamma_n(a) \frac{\log^m x}{x} - \gamma_m(a) \frac{\log^n(a-x)}{a-x} \right\} dx \\ & + \int_a^1 \gamma_m(x) \gamma_n(1+a-x) dx \\ & + \frac{\gamma_n(a)}{m+1} \log^{m+1} a + \frac{\gamma_m(a)}{n+1} \log^{n+1} a. \end{aligned} \quad (3.25)$$

Proposition 3.4 (Ordinary integral realization). *Both integrals in (3.25) converge absolutely, and*

$$\mathcal{I}_{m,n}(a) = (T_m * T_n)(a), \quad 0 < a < 1.$$

Consequently,

$$\mathcal{I}_{0,0}(a) = 2\gamma_1(a) + \zeta(2), \quad (3.26)$$

$$\mathcal{I}_{0,1}(a) = \frac{3}{2}\gamma_2(a) - \zeta(2)\gamma_0(a) - \zeta(3), \quad (3.27)$$

$$\mathcal{I}_{0,2}(a) = \frac{4}{3}\gamma_3(a) - 2\zeta(2)\gamma_1(a) + 2\zeta(3)\gamma_0(a) + 2\zeta(4), \quad (3.28)$$

$$\mathcal{I}_{1,1}(a) = \gamma_3(a) - 2\zeta(2)\gamma_1(a) + 2\zeta(3)\gamma_0(a) + \frac{1}{4}\zeta(4). \quad (3.29)$$

Proof. Near $x = 0$, the shift formula for γ_m and smoothness of γ_n near a show that the first subtracted integrand is $O(1 + |\log x|^m)$. The same argument applies near $x = a$. The second integral has no singular endpoints for fixed $a \in (0, 1)$.

For a direct identification, first restrict a to a compact subinterval of $(0, 1)$. The singular supports of $T_m(x)$ and $T_n(a-x)$ are separated in the integration variable. A smooth partition of unity around $x = 0$, $x = a$ and their complement therefore defines their product: in each singular neighborhood only one factor is a distribution and the other is smooth. Applying the two cutoff formulas (2.11), and then combining the complementary integrals, gives

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^{a-\epsilon} \gamma_m(x) \gamma_n(a-x) dx + \int_a^1 \gamma_m(x) \gamma_n(1+a-x) dx \right. \\ \left. + \frac{\gamma_n(a)}{m+1} \log^{m+1} \epsilon + \frac{\gamma_m(a)}{n+1} \log^{n+1} \epsilon \right\}. \end{aligned}$$

The use of cutoff singular coefficients at a introduces errors at most $O(\epsilon |\log \epsilon|^M)$ for a fixed M , hence no additional constant. Integrating the explicitly subtracted logarithmic terms evaluates this limit as (3.25). This agrees with convolution defined by addition on the torus. Equations (3.26)–(3.29) follow by restricting (3.10)–(3.13) to $(0, 1)$, where Δ is the constant -1 . $\square$

In terms of the ordinary digamma function, the first identity reads

$$\begin{aligned} 2\gamma_1(a) + \zeta(2) = & \int_0^a \left\{ \psi(x) \psi(a-x) + \psi(a) \left(\frac{1}{x} + \frac{1}{a-x} \right) \right\} dx \\ & + \int_a^1 \psi(x) \psi(1+a-x) dx - 2\psi(a) \log a. \end{aligned} \quad (3.30)$$

No divergent integral occurs on the right: the braces are to be combined before integration. For instance,

$$\mathcal{I}_{0,0}(1/2) = 2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 + \zeta(2),$$

because $\gamma_1(1/2) = \gamma_1 - 2\gamma \log 2 - \log^2 2$. The factor of the unshifted γ_1 in the latter identity is 1, as follows directly from $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$.

3.5 Every repeated Stieltjes primitive in one finite formula

For a zero-mean periodic distribution define D^{-q} by its Fourier multiplier $(2\pi in)^{-q}$ at $n \neq 0$ and zero at $n = 0$. This is the unique repeated primitive having zero mean at each stage; it equals convolution with W_{1-q} . Define the rational coefficients

$$\prod_{r=1}^{q-1} \left(1 - \frac{w}{r}\right)^{-1} = \sum_{j \geq 0} h_j^{(q-1)} w^j, \quad h_0^{(0)} = 1, \quad h_j^{(0)} = 0 \quad (j > 0). \quad (3.31)$$

They are complete homogeneous symmetric functions in $1, 1/2, \dots, 1/(q-1)$; equivalently their generating function is

$$\exp \left(\sum_{r \geq 1} \frac{H_{q-1}^{(r)}}{r} w^r \right), \quad H_N^{(r)} = \sum_{j=1}^N \frac{1}{j^r}.$$

For example $h_1^{(N)} = H_N$ and $h_2^{(N)} = (H_N^2 + H_N^{(2)})/2$.

Theorem 3.5 (All repeated primitives and their constants). *For $q \geq 1, n \geq 0$, the zero-mean q -fold primitive of T_n is*

$$D^{-q}T_n(x) = \frac{(-1)^{n+1}n!}{(q-1)!} \sum_{j=0}^{n+1} \frac{h_{n+1-j}^{(q-1)}}{j!} \zeta^{[j]}(1-q, x), \quad 0 < x < 1. \quad (3.32)$$

Here $\zeta^{[j]}$ means the j th derivative in the first argument. The right side is periodized as a locally integrable function. For $q \geq 2$ it is continuous, with the regularity implied by its endpoint terms $x^{q-1} \log^j x$.

If $q > k \geq 0$ and $p = q - k$, then

$$D^{-q}P_k(x) = \frac{\zeta'(1-p, x) + (H_{p-1} - H_k)\zeta(1-p, x)}{(p-1)!}, \quad (3.33)$$

$$D^{-q}Q_k(x) = \frac{\zeta'(1-p, x) + H_{p-1}\zeta(1-p, x)}{(p-1)!}. \quad (3.34)$$

Proof. The normalized family gives, identically in w near zero,

$$D^{-q}Z_{1+w} = \Gamma(-w)W_{1-q+w} = \frac{Z_{1-q+w}}{(-w)_q}. \quad (3.35)$$

Since

$$\frac{1}{(-w)_q} = -\frac{1}{(q-1)!w} \prod_{r=1}^{q-1} (1 - w/r)^{-1},$$

the coefficient $(-1)^n n! [w^n]$ is exactly (3.32). For $q \geq 1$ the relevant Hurwitz jets have order $1 - q \leq 0$ and are locally integrable. Their means vanish, by the integrable Hurwitz identity

continued and differentiated below 1, so no hidden constants occur. For $q \geq 2$, the shift formula isolates the only nonanalytic endpoint term as a finite combination of $x^{q-1} \log^j x$; it tends to zero and matches the finite endpoint value at 1.

For (3.33), the Fourier coefficient after taking q primitives is $(2\pi i n)^{-p}(\gamma + L_n - H_k)$. Differentiation of $Z_s = \Gamma(1-s)W_s$ at $s = 1-p$ gives

$$\frac{Z'_{1-p}}{(p-1)!} = \partial_s W_s|_{s=1-p} - \psi(p)W_{1-p}.$$

Use $\psi(p) = H_{p-1} - \gamma$. Adding the primitive of $H_k \delta_0^{(k)}$ gives (3.34). $\square$

At $q = 1$, (3.32) becomes the familiar Stieltjes primitive $(-1)^{n+1} \zeta^{[n+1]}(0, x)/(n+1)$. At every larger q it gives all lower jets and all constants of integration explicitly. For example,

$$\begin{aligned} D^{-2}T_1(x) &= \frac{1}{2}\zeta''(-1, x) + \zeta'(-1, x) + \zeta(-1, x), \\ D^{-3}T_1(x) &= \frac{1}{4}\zeta''(-2, x) + \frac{3}{4}\zeta'(-2, x) + \frac{7}{8}\zeta(-2, x). \end{aligned}$$

The family in (3.34) is the classical balanced negative polygamma family. Equation (3.33) records the correction required when one starts from spectral finite parts instead of derivatives of periodic $\log \Gamma$.

3.6 A convergent Gamma convolution as a normalization check

One can remove the singularities by integrating. In particular, for $g = \log \Gamma - \frac{1}{2} \log(2\pi)$,

$$\begin{aligned} \int_0^a g(x)g(a-x) dx + \int_a^1 g(x)g(1+a-x) dx \\ = \zeta''(-1, a) + 2\zeta'(-1, a) + (2 - \zeta(2))\zeta(-1, a), \quad 0 < a \leq 1. \end{aligned} \quad (3.36)$$

Indeed $Dg = P_0$, so $g * g = D^{-2}(P_0 * P_0)$; apply (3.10) and (3.32). The right side extends continuously to the endpoints. This is a differentiated instance of Sun's classical ordinary convolution, included to exhibit the agreement of the singular calculus with convergent Gamma integrals.

4 Nonintegral twists: Lerch jets, contact terms, and the half-twist

The pinned convolution-algebra continuation explicitly asks for a finite-part Stieltjes normal form for nonintegral frequency shifts, with an all-index half-twist multiplication table that includes its point-supported counterterms [3, "Twists, characters, and alternate endpoint normalizations"]. We give such a normalization and table here. The underlying Lerch function is classical [11, §25.14]; at the half-twist its order jets are the modified Stieltjes constants of Hu and Kim [20]. An ordinary, nonsingular alternating Hurwitz convolution is already given by Zhu, Hu, and Kim [21, Theorem 1.13]. The additional content below is the continuation through the singular orders, including the exact contact terms, covariant primitives, and degeneration to the untwisted normalization. In particular, twisting changes the special-function jets and the convolution identity; the positive-integer zeta values in the Gamma correction remain the same.

4.1 The meromorphic family and its endpoint normalization

Fix $0 < \theta < 1$, put $\omega = 2\pi\theta$ and $z = e^{-i\omega}$, and retain the Fourier and convolution conventions of Section 2. Any other real nonintegral twist $\theta = m + \vartheta$, $m \in \mathbb{Z}$, $0 < \vartheta < 1$, is obtained by the periodic modulation $U(x) \mapsto e^{-2\pi i m x} U(x)$. Thus the chosen interval loses no real nonintegral shifts. Define

$$\lambda_{n,\theta} = 2\pi i(n + \theta), \quad L_{n,\theta} = \log(2\pi|n + \theta|) + \frac{i\pi}{2} \operatorname{sgn}(n + \theta), \quad n \in \mathbb{Z}. \quad (4.1)$$

Every power of $\lambda_{n,\theta}$ uses this logarithm. There is no zero frequency to remove. We write

$$D_\theta = D + 2\pi i\theta, \quad \widehat{D_\theta U}(n) = \lambda_{n,\theta} \widehat{U}(n).$$

Here D_θ acts on ordinary periodic distributions, rather than on functions satisfying a different boundary condition. The nonperiodic factor $e^{-i\omega x}$ below is used only for ordinary representatives on $0 < x < 1$ and for local endpoint Taylor coefficients. It is not used as a global multiplier on periodic distributions; the Fourier or cutoff definitions fix those extensions. Explicitly, for $m \in \mathbb{Z}$ the extension to shifted intervals is $W_s^{\theta+m}(x) = e^{-2\pi i m x} W_s^\theta(x)$, with the same identity for Z_s , U_n , and P_k after their definitions below.

Theorem 4.1 (Twisted Lerch family). *The Fourier coefficients*

$$\widehat{W_s^\theta}(n) = \lambda_{n,\theta}^{s-1}, \quad n \in \mathbb{Z}, \quad (4.2)$$

define an entire distribution-valued function of s . The family

$$Z_s^\theta = \Gamma(1-s) W_s^\theta \quad (4.3)$$

is meromorphic, with simple poles precisely at $s = k + 1$, $k \geq 0$, and

$$\operatorname{Res}_{s=k+1} Z_s^\theta = \frac{(-1)^{k+1}}{k!} D_\theta^k \delta_0. \quad (4.4)$$

For $\operatorname{Re} s < 1$, it is the locally integrable periodic extension of

$$f_s^\theta(x) = e^{-i\omega x} \Phi(z, s, x), \quad 0 < x < 1. \quad (4.5)$$

For every s , the restriction of the meromorphic distribution to $(0, 1)$ is the entire function of s on the right of (4.5). The convolution and differential identities are

$$W_s^\theta * W_t^\theta = W_{s+t-1}^\theta, \quad W_1^\theta = \delta_0, \quad (4.6)$$

$$D_\theta W_s^\theta = W_{s+1}^\theta, \quad D_\theta Z_s^\theta = -s Z_{s+1}^\theta. \quad (4.7)$$

Consequently,

$$Z_s^\theta * Z_t^\theta = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} Z_{s+t-1}^\theta \quad (4.8)$$

as a meromorphic identity, interpreted through (4.6) at intersections of polar divisors.

Proof. On compact sets of s , the coefficients in (4.2), and each fixed number of their order derivatives, have a common polynomial growth bound in $|n|$. Pairing their Fourier series with the rapidly decreasing Fourier coefficients of a smooth test function therefore gives normal convergence. This proves entire dependence. The Fourier multiplier of $D_\theta^k \delta_0$ is $\lambda_{n,\theta}^k$; the residues of $\Gamma(1-s)$ prove (4.4). These distributions are nonzero, so every displayed pole is present.

For completeness, the Lerch identification can be proved without assuming a boundary-value Fourier formula. Initially let $0 < \operatorname{Re} s < 1$ and $\epsilon > 0$. Abel damping gives

$$f_{s,\epsilon}^\theta(x) = \sum_{m \geq 0} e^{-(\epsilon+i\omega)(m+x)} (m+x)^{-s}.$$

Absolute convergence permits unfolding its n th Fourier coefficient:

$$\begin{aligned} \int_0^1 f_{s,\epsilon}^\theta(x) e^{-2\pi i n x} dx &= \int_0^\infty t^{-s} e^{-(\epsilon+\lambda_{n,\theta})t} dt \\ &= \Gamma(1-s)(\epsilon+\lambda_{n,\theta})^{s-1}. \end{aligned}$$

Since $z \neq 1$, the partial sums of z^m are bounded. Summation by parts shows that the tail starting at $m = 1$ converges uniformly in $x \in [0, 1]$, and locally uniformly for s in this strip, as $\epsilon \downarrow 0$. The remaining term is bounded by an integrable multiple of $x^{-\operatorname{Re} s}$. Thus the limit can be taken in $L^1(0, 1)$ and in its Fourier coefficients, giving (4.5). Repeated summation by parts, or finite differences of $(m+x)^{-s}$, continues the Lerch tail holomorphically to every s , locally uniformly when x stays away from zero. Near zero its only nonsmooth term is

$$f_s^\theta(x) = e^{-i\omega x} x^{-s} + e^{-i\omega x} z \Phi(z, s, 1+x). \quad (4.9)$$

This proves L^1 holomorphy throughout $\operatorname{Re} s < 1$ and the asserted continuation away from the endpoint. The remaining identities follow by multiplying the Fourier coefficients; the Gamma recurrence gives the last identity of (4.7). $\square$

The pointwise function in (4.5) has no pole in s . Every pole in (4.3) is an endpoint distribution. This is different from the untwisted Hurwitz family, whose pole at $s = 1$ also has an ordinary constant contribution away from the endpoint.

For $n \geq 0$ define ordinary Lerch jets and their periodic finite parts by

$$c_n^\theta(x) = (-1)^n \partial_s^n f_s^\theta(x) \Big|_{s=1}, \quad 0 < x < 1, \quad (4.10)$$

$$Z_{1+u}^\theta = -\frac{\delta_0}{u} + \sum_{n \geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \quad (4.11)$$

Set, also,

$$P_k^\theta = (-1)^{k+1} k! \operatorname{FP}_{s=k+1} Z_s^\theta, \quad k \geq 0. \quad (4.12)$$

In particular $P_0^\theta = -U_0^\theta$. Off the endpoint it equals $p_k^\theta(x) = (-1)^{k+1} k! f_{k+1}^\theta(x)$.

Proposition 4.2 (Explicit singular normalization). *For a smooth periodic test function ϕ , put $h(x) = e^{-i\omega x} \phi(x)$ near zero. Then*

$$\langle U_n^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 c_n^\theta(x) \phi(x) dx + \frac{\phi(0)}{n+1} \log^{n+1} \epsilon \right\}. \quad (4.13)$$

If $F_k^\theta = \text{FP}_{s=k+1} Z_s^\theta$ and $k \geq 1$, then

$$\langle F_k^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 f_{k+1}^\theta(x) \phi(x) dx - \sum_{j=0}^{k-1} \frac{h^{(j)}(0)}{j!} \frac{\epsilon^{j-k}}{k-j} + \frac{h^{(k)}(0)}{k!} \log \epsilon \right\}. \quad (4.14)$$

These normalizations retain the zero Fourier coefficient; no mean-zero condition is imposed.

Proof. Write the second term in (4.9) as $r_s^\theta(x)$. After subtracting the degree- N Taylor polynomial of h , the continuation of the test-function pairing is

$$\begin{aligned} \langle Z_s^\theta, \phi \rangle &= \int_0^1 x^{-s} \left(h(x) - \sum_{j=0}^N \frac{h^{(j)}(0)}{j!} x^j \right) dx + \sum_{j=0}^N \frac{h^{(j)}(0)}{j!(j+1-s)} \\ &\quad + \int_0^1 r_s^\theta(x) \phi(x) dx. \end{aligned} \quad (4.15)$$

The first integral is holomorphic for $\text{Re } s < N+2$, and the last integral is entire. At $s = 1+u$, expand $x^{-1-u} = x^{-1} \sum_n (-1)^n u^n \log^n x / n!$ after the constant subtraction. This gives (4.13). At $s = k+1$, take the constant Laurent coefficient and integrate each subtracted monomial explicitly; this gives (4.14). The residue computed this way agrees with (4.4), because

$$\langle D_\theta^k \delta_0, \phi \rangle = (-1)^k h^{(k)}(0).$$

This also fixes the signs of all the lower derivatives of δ_0 inside $D_\theta^k \delta_0$. □

4.2 All-index multiplication and covariant contact terms

Let

$$B_\theta(u) = -u Z_{1+u}^\theta = \Gamma(1-u) W_{1+u}^\theta, \quad G_\theta(u) = \sum_{n \geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \quad (4.16)$$

Thus $B_\theta(u) = \delta_0 - u G_\theta(u)$. All the following equalities are analytic germs at zero in the algebra of periodic distributions with its usual convolution identity δ_0 .

Theorem 4.3 (Twisted Stieltjes normal form). *With $C(u, v)$ as in (3.5),*

$$B_\theta(u) * B_\theta(v) = C(u, v) B_\theta(u+v), \quad (4.17)$$

$$B_\theta(u) = \exp_* \left(u P_0^\theta + \sum_{r \geq 2} \frac{\zeta(r)}{r} u^r \delta_0 \right). \quad (4.18)$$

In particular,

$$U_n^\theta = \frac{(-1)^{n+1}}{n+1} \mathcal{B}_{n+1}^* (P_0^\theta, \zeta(2) \delta_0, 2! \zeta(3) \delta_0, \dots, n! \zeta(n+1) \delta_0). \quad (4.19)$$

An explicit all-index multiplication table is furnished by the removable coefficient formula

$$\begin{aligned} G_\theta(u) * G_\theta(v) &= \frac{1}{uv} \{ u G_\theta(u) + v G_\theta(v) - (u+v) C(u, v) G_\theta(u+v) \\ &\quad + (C(u, v) - 1) \delta_0 \}. \end{aligned} \quad (4.20)$$

More precisely, $U_m^\theta * U_n^\theta$ is $(-1)^{m+n} m! n!$ times the coefficient of $u^m v^n$ on the right. This is a finite combination of $\delta_0, U_0^\theta, \dots, U_{m+n+1}^\theta$, with coefficients polynomial in ordinary zeta values.

Proof. The first identity follows from (4.6). At every Fourier mode, including $n = 0$,

$$\widehat{P}_0^\theta(n) = \gamma + L_{n,\theta}, \quad \widehat{B_\theta(u)}(n) = \exp \left((\gamma + L_{n,\theta})u + \sum_{r \geq 2} \frac{\zeta(r)}{r} u^r \right).$$

This proves (4.18) and the Bell formula, with normal convergence after test-function pairing. Substitute $B_\theta(u) = \delta_0 - uG_\theta(u)$ into (4.17) and solve for $G_\theta(u) * G_\theta(v)$ to obtain (4.20). Its numerator vanishes when either variable is zero, so the quotient is removable. Extracting a fixed coefficient uses only finitely many coefficients of the Gamma quotient and of G_θ . $\square$

For example, the full identities, including their contact masses, are

$$U_0^\theta * U_0^\theta = 2U_1^\theta - \zeta(2)\delta_0, \quad (4.21)$$

$$U_0^\theta * U_1^\theta = \frac{3}{2}U_2^\theta - \zeta(2)U_0^\theta + \zeta(3)\delta_0, \quad (4.22)$$

$$U_0^\theta * U_2^\theta = \frac{4}{3}U_3^\theta - 2\zeta(2)U_1^\theta + 2\zeta(3)U_0^\theta - 2\zeta(4)\delta_0, \quad (4.23)$$

$$U_1^\theta * U_1^\theta = U_3^\theta - 2\zeta(2)U_1^\theta + 2\zeta(3)U_0^\theta - \frac{1}{4}\zeta(4)\delta_0. \quad (4.24)$$

The presence of δ_0 , rather than $\delta_0 - 1$, matters even when these formulas are eventually evaluated away from zero.

Theorem 4.4 (All covariant contact corrections). *For $k \geq 0$, with $H_0 = 0$,*

$$\widehat{P}_k^\theta(n) = \lambda_{n,\theta}^k (\gamma + L_{n,\theta} - H_k), \quad (4.25)$$

$$D_\theta P_k^\theta = P_{k+1}^\theta + \frac{1}{k+1} D_\theta^{k+1} \delta_0, \quad (4.26)$$

$$Q_k^\theta := D_\theta^k P_0^\theta = P_k^\theta + H_k D_\theta^k \delta_0. \quad (4.27)$$

Hence the all-index twisted polygamma product is

$$P_k^\theta * P_\ell^\theta = D_\theta^{k+\ell} \{ 2U_1^\theta + (H_k + H_\ell)U_0^\theta + (H_k H_\ell - \zeta(2))\delta_0 \}. \quad (4.28)$$

More generally, let $R_{n,k}^\theta$ be the raw Hadamard finite part of the ordinary function $D_\theta^k c_n^\theta$ on $(0,1)$, using Taylor subtraction of its powers of x and $\log x$ at zero and no additional endpoint mass. Then

$$D_\theta^k U_n^\theta = R_{n,k}^\theta - h_{k,n} D_\theta^k \delta_0, \quad h_{k,n} = (-1)^n n! [u^{n+1}] \prod_{j=1}^k \left(1 + \frac{u}{j} \right). \quad (4.29)$$

In particular $h_{k,0} = H_k$, and $h_{k,n} = 0$ for $n \geq k$.

Proof. At $s = k + 1 + u$,

$$\Gamma(-k-u) = \frac{(-1)^{k+1}}{k!u} + \frac{(-1)^k}{k!} (H_k - \gamma) + O(u).$$

Multiply this by $\lambda_{n,\theta}^k e^{uL_{n,\theta}}$ and take the finite part. This proves (4.25). The two derivative formulas follow by multiplication by $\lambda_{n,\theta}$ and $H_{k+1} - H_k = 1/(k+1)$. Multiplying the two affine polynomials $\gamma + L - H_k$ and $\gamma + L - H_\ell$ and using (4.21) gives (4.28).

For clarity, the raw finite part in (4.29) can equivalently be defined by the coefficientwise constant cutoff term of $D_\theta^k f_{1+u}^\theta$. The Mellin subtraction (4.15) identifies the regular Laurent

coefficients of Z_{k+1+u}^θ with those cutoff finite parts of its order derivatives. Repeated application of (4.7) gives

$$D_\theta^k Z_{1+u}^\theta = (-1)^k (1+u)_k Z_{k+1+u}^\theta.$$

The contribution from the pole on the right is exactly

$$-\frac{1}{u} \prod_{j=1}^k \left(1 + \frac{u}{j}\right) D_\theta^k \delta_0.$$

Its constant and higher coefficients supply all the difference between derivative-compatible and raw finite parts. Equating the coefficients of $(-1)^n u^n / n!$ proves (4.29). $\square$

For example, $D_\theta^2 \delta_0 = \delta_0'' + 4\pi i \theta \delta_0' - (2\pi\theta)^2 \delta_0$. Thus (4.27) specifies the lower point-supported terms as well as the highest derivative. Writing only $H_k \delta_0^{(k)}$ would give the wrong normalization for a nonzero twist.

4.3 Rational twists and ordinary special functions

Proposition 4.5 (Finite Hurwitz reduction). *Let $\theta = p/q$, where $1 \leq p < q$ are relatively prime, and put $z = e^{-2\pi i p/q}$. Then*

$$f_s^{p/q}(x) = \frac{e^{-2\pi i p x/q}}{q^s} \sum_{r=0}^{q-1} z^r \zeta\left(s, \frac{x+r}{q}\right), \quad 0 < x < 1. \quad (4.30)$$

The sum is entire in s : its Hurwitz residues cancel. Consequently,

$$c_n^{p/q}(x) = \frac{e^{-2\pi i p x/q}}{q} \sum_{r=0}^{q-1} z^r \sum_{j=0}^n \binom{n}{j} (\log q)^{n-j} \gamma_j \left(\frac{x+r}{q}\right), \quad (4.31)$$

$$p_k^{p/q}(x) = \frac{e^{-2\pi i p x/q}}{q^{k+1}} \sum_{r=0}^{q-1} z^r \psi^{(k)}\left(\frac{x+r}{q}\right), \quad k \geq 0. \quad (4.32)$$

These ordinary functions, extended by (4.13) and (4.14), give the finite-part distributions in Theorems 4.3 and 4.4.

Proof. For $\operatorname{Re} s > 1$, split the defining Lerch series into its residue classes modulo q . This gives (4.30); analytic continuation proves it everywhere. Since $\sum_r z^r = 0$, the common Hurwitz pole vanishes before the factor q^{-s} is expanded. Differentiating the resulting Taylor expansion at $s = 1$ gives (4.31). At positive integers use the Hurwitz–polygamma identity; its limiting case at $s = 1$ uses $\gamma_0 = -\psi$ and the same pole cancellation. This proves (4.32) also for $k = 0$. $\square$

Thus additive rational twists require finite colored combinations of Hurwitz jets, while the coefficients governing their convolution remain the ordinary $\zeta(r)$ from $\log \Gamma(1-u)$. No replacement by Dirichlet eta values is needed in (4.19) or (4.20).

4.4 The half-twist and convergent alternating identities

At $\theta = \frac{1}{2}$, set

$$\zeta_E(s, x) = \Phi(-1, s, x) = 2^{-s} \left\{ \zeta\left(s, \frac{x}{2}\right) - \zeta\left(s, \frac{x+1}{2}\right) \right\}, \quad (4.33)$$

$$\tilde{\gamma}_n(x) = (-1)^n \partial_s^n \zeta_E(s, x)|_{s=1}. \quad (4.34)$$

These modified Stieltjes constants are the established functions of Hu and Kim [20]. The explicit conversion is

$$\tilde{\gamma}_n(x) = \frac{1}{2} \sum_{j=0}^n \binom{n}{j} (\log 2)^{n-j} \left\{ \gamma_j\left(\frac{x}{2}\right) - \gamma_j\left(\frac{x+1}{2}\right) \right\}, \quad c_n^{1/2}(x) = e^{-\pi i x} \tilde{\gamma}_n(x). \quad (4.35)$$

In particular, the Nielsen beta function is

$$\beta(x) = \tilde{\gamma}_0(x) = \frac{1}{2} \left\{ \psi\left(\frac{x+1}{2}\right) - \psi\left(\frac{x}{2}\right) \right\}. \quad (4.36)$$

The elementary shift relation

$$\tilde{\gamma}_n(x) + \tilde{\gamma}_n(x+1) = \frac{\log^n x}{x} \quad (4.37)$$

fixes its endpoint singularity, including every logarithmic order.

Theorem 4.6 (All-index alternating finite-part convolution). *For $0 < a < 1$ and $m, n \geq 0$, the following expression consists of absolutely convergent integrals:*

$$\begin{aligned} \mathcal{A}_{mn}(a) = & \int_0^a \left\{ \tilde{\gamma}_m(x) \tilde{\gamma}_n(a-x) - \tilde{\gamma}_n(a) \frac{\log^m x}{x} - \tilde{\gamma}_m(a) \frac{\log^n(a-x)}{a-x} \right\} dx \\ & - \int_a^1 \tilde{\gamma}_m(x) \tilde{\gamma}_n(1+a-x) dx \\ & + \frac{\tilde{\gamma}_n(a)}{m+1} \log^{m+1} a + \frac{\tilde{\gamma}_m(a)}{n+1} \log^{n+1} a. \end{aligned} \quad (4.38)$$

It equals $e^{\pi i a} (U_m^{1/2} * U_n^{1/2})(a)$, where the convolution is smooth on $(0, 1)$. Every $\mathcal{A}_{mn}$ is therefore the finite linear combination of $\tilde{\gamma}_0, \dots, \tilde{\gamma}_{m+n+1}$ obtained from (4.20) by discarding its δ_0 term. In particular,

$$\mathcal{A}_{00}(a) = 2\tilde{\gamma}_1(a), \quad (4.39)$$

$$\mathcal{A}_{01}(a) = \frac{3}{2}\tilde{\gamma}_2(a) - \zeta(2)\tilde{\gamma}_0(a), \quad (4.40)$$

$$\mathcal{A}_{11}(a) = \tilde{\gamma}_3(a) - 2\zeta(2)\tilde{\gamma}_1(a) + 2\zeta(3)\tilde{\gamma}_0(a). \quad (4.41)$$

Proof. For integrable orders, convolution on the circle splits into $0 < x < a$ and $a < x < 1$. The phases of the two factors have product $e^{-\pi i a}$ in the first interval and $e^{-\pi i(1+a)} = -e^{-\pi i a}$ in the second. This is the source of the alternating sign in (4.38), and agrees with the nonsingular alternating convolution of [21, Theorem 1.13].

To continue to the jets, use the normalized cutoff distributions in (4.13). On a compact subinterval of $0 < a < 1$, the two endpoint singularities in the first integral occur at distinct points, $x = 0$ and $x = a$. Their only divergent terms are the two explicit subtractions in (4.38). For instance, at zero, (4.37) writes the first factor as $\log^m x/x$ plus a smooth function, while

$\tilde{\gamma}_n(a-x) - \tilde{\gamma}_n(a) = O(x)$. The subtracted integrand is therefore $O(1 + |\log x|^m)$ there, and has the analogous integrable behavior at a . The wrapped integral is smooth. Integration of the two subtracted singular monomials gives exactly the two finite $\log a$ terms shown. This verifies the cutoff expression and permits all parameter coefficients to be taken under the subtracted integrals. Equivalently it is the restriction of the meromorphic convolution identity (4.8). The identities (4.39)–(4.41) now follow from (4.21), (4.22), and (4.24). $\square$

For ordinary digamma functions this gives the concrete identity

$$\begin{aligned} \int_0^a \left\{ \beta(x)\beta(a-x) - \beta(a) \left(\frac{1}{x} + \frac{1}{a-x} \right) \right\} dx - \int_a^1 \beta(x)\beta(1+a-x) dx + 2\beta(a) \log a \\ = \gamma_1 \left(\frac{a}{2} \right) - \gamma_1 \left(\frac{a+1}{2} \right) + 2 \log 2 \beta(a). \end{aligned} \quad (4.42)$$

The full distributional identity behind it still contains $-\zeta(2)\delta_0$. Omitting that mass would preserve (4.42) on $0 < a < 1$ but would destroy the Fourier coefficients and every identity involving endpoint tests.

Corollary 4.7 (A quarter-Gamma evaluation). *At $a = \frac{1}{2}$, the absolutely convergent integral in (4.42) has the exact value*

$$\mathcal{A}_{00} \left(\frac{1}{2} \right) = \gamma_1 \left(\frac{1}{4} \right) - \gamma_1 \left(\frac{3}{4} \right) + \pi \log 2 = \pi \left\{ 4 \log \Gamma \left(\frac{1}{4} \right) - \gamma - 3 \log(2\pi) \right\}. \quad (4.43)$$

Proof. Distinguish the Dirichlet beta function $\beta_D(s) = \sum_{r \geq 0} (-1)^r (2r+1)^{-s}$ from the Nielsen function $\beta(x)$ in (4.36). The identity $\zeta_E(s, \frac{1}{2}) = 2^s \beta_D(s)$ gives

$$2\tilde{\gamma}_1 \left(\frac{1}{2} \right) = -\pi \log 2 - 4\beta'_D(1),$$

since $\beta_D(1) = \pi/4$. The completed Dirichlet functional equation [12, §25.15(i)] is, for this character,

$$\Lambda(s) = \left(\frac{4}{\pi} \right)^{(s+1)/2} \Gamma \left(\frac{s+1}{2} \right) \beta_D(s), \quad \Lambda(s) = \Lambda(1-s).$$

Also

$$\beta_D(s) = 4^{-s} \left\{ \zeta \left(s, \frac{1}{4} \right) - \zeta \left(s, \frac{3}{4} \right) \right\}, \quad \beta_D(0) = \frac{1}{2}, \quad \beta'_D(0) = \log \Gamma \left(\frac{1}{4} \right) - \log \Gamma \left(\frac{3}{4} \right) - \log 2.$$

Logarithmic differentiation of the completed functional equation at 0 and 1, followed by the Gamma reflection formula, yields

$$\beta'_D(1) = \frac{\pi}{4} \left\{ \gamma + 2 \log 2 + 3 \log \pi - 4 \log \Gamma \left(\frac{1}{4} \right) \right\}.$$

Substitution proves (4.43); the first equality also uses $\beta(\frac{1}{2}) = \pi/2$ in (4.42). $\square$

4.5 Reflection at the half-twist

For the half-twist, the map $n \mapsto -n-1$ reverses the shifted frequency $n + \frac{1}{2}$. Its periodic realization is

$$\mathcal{J}U(x) = e^{-2\pi i x} U(-x), \quad \widehat{\mathcal{J}U}(n) = \widehat{U}(-n-1). \quad (4.44)$$

This is an involutive convolution automorphism with $\mathcal{J}\delta_0 = \delta_0$ and $D_{1/2}\mathcal{J} = -\mathcal{J}D_{1/2}$. Ordinary reflection alone does not reverse the half-shifted frequencies. Also define

$$\widehat{\mathcal{H}_{1/2}U}(n) = -i \operatorname{sgn}\left(n + \frac{1}{2}\right) \widehat{U}(n), \quad n \in \mathbb{Z}.$$

Here $\mathcal{H}_{1/2}^2 = -\operatorname{Id}$ on all periodic distributions, because every shifted frequency is nonzero.

Corollary 4.8 (Reflected half-twist table). *Writing $U_n = U_n^{1/2}$ and $B = B_{1/2}$ in this corollary,*

$$B(u) * \mathcal{J}B(v) = C(u, v) (\cos(\pi v) \operatorname{Id} + \sin(\pi v) \mathcal{H}_{1/2}) B(u + v). \quad (4.45)$$

Thus the coefficient prescription gives an all-index reflected table as well. Two explicit instances are

$$U_0 * \mathcal{J}U_1 = \frac{1}{2}U_2 + \mathcal{J}U_2 + \zeta(2)(U_0 + \mathcal{J}U_0) + \zeta(3)\delta_0, \quad (4.46)$$

$$U_1 * \mathcal{J}U_1 = \frac{1}{2}(U_3 + \mathcal{J}U_3) + 2\zeta(2)(U_1 + \mathcal{J}U_1) + \zeta(3)(U_0 + \mathcal{J}U_0) + \frac{7}{2}\zeta(4)\delta_0. \quad (4.47)$$

For the polygamma finite parts,

$$P_k^{1/2} * \mathcal{J}P_\ell^{1/2} = (-1)^\ell D_{1/2}^{k+\ell} \{U_1 + \mathcal{J}U_1 + H_\ell U_0 + H_k \mathcal{J}U_0 + (H_k H_\ell + 2\zeta(2))\delta_0\}. \quad (4.48)$$

Proof. At mode n put $\sigma = \operatorname{sgn}(n + \frac{1}{2})$. The logarithms of $\lambda_{n,1/2}$ and $-\lambda_{n,1/2}$ differ by $i\pi\sigma$. Dividing the Fourier coefficient of the left side of (4.45) by that of $B(u + v)$ gives $C(u, v)e^{-i\pi v\sigma}$, which is exactly the multiplier on the right. Coefficient extraction, with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$, proves the two displayed instances. Multiplication of $\gamma + L_{n,1/2} - H_k$ and $\gamma + L_{-n-1,1/2} - H_\ell$ proves (4.48); the factor $(-1)^\ell$ comes from $\lambda_{-n-1,1/2}^\ell = (-\lambda_{n,1/2})^\ell$. $\square$

The modulation in (4.44) is periodic specifically because $2\theta = 1$. For a general twist, frequency reversal connects θ with $1 - \theta$; the same fixed-twist involution should not be asserted without changing the space or the twist.

4.6 Unique covariant primitives

Because $\lambda_{n,\theta} \neq 0$ for every n , the operator D_θ has a unique inverse on periodic distributions:

$$\widehat{D_\theta^{-q}U}(n) = \lambda_{n,\theta}^{-q} \widehat{U}(n), \quad q \geq 1.$$

This removes the choice of an integration constant that occurs for ordinary periodic primitives.

Theorem 4.9 (All-order covariant antiderivatives). *Define the complete homogeneous reciprocal-harmonic coefficients by*

$$\sum_{j \geq 0} h_j^{(q-1)} w^j = \prod_{r=1}^{q-1} \left(1 - \frac{w}{r}\right)^{-1}; \quad h_0^{(0)} = 1, \quad h_j^{(0)} = 0 \ (j > 0). \quad (4.49)$$

Then for $q \geq 1$ and $n \geq 0$,

$$D_\theta^{-q}U_n^\theta = \frac{(-1)^{n+1}n!}{(q-1)!} \sum_{j=0}^{n+1} \frac{h_{n+1-j}^{(q-1)}}{j!} \partial_s^j Z_s^\theta \Big|_{s=1-q}. \quad (4.50)$$

Its ordinary value on $(0, 1)$ is obtained by replacing Z_s^θ by f_s^θ . For $q \geq 2$ this is a periodic C^{q-2} function. For $q > k \geq 0$, setting $p = q - k$ gives

$$D_\theta^{-q} P_k^\theta = \frac{\partial_s Z_s^\theta|_{s=1-p} + (H_{p-1} - H_k) Z_{1-p}^\theta}{(p-1)!}, \quad (4.51)$$

$$D_\theta^{-q} Q_k^\theta = \frac{\partial_s Z_s^\theta|_{s=1-p} + H_{p-1} Z_{1-p}^\theta}{(p-1)!}. \quad (4.52)$$

At the half-twist, the first primitive has the explicit Gamma form

$$D_{1/2}^{-1} U_0^{1/2}(x) = e^{-\pi i x} \left\{ \log \Gamma \left(\frac{x+1}{2} \right) - \log \Gamma \left(\frac{x}{2} \right) + \frac{1}{2} \log 2 \right\}, \quad 0 < x < 1. \quad (4.53)$$

The displayed additive term is fixed by the periodic distributional normalization.

Proof. The spectral definition and the Gamma recurrence give

$$D_\theta^{-q} Z_{1+u}^\theta = \frac{Z_{1-q+u}^\theta}{(-u)_q}, \quad \frac{1}{(-u)_q} = -\frac{1}{(q-1)!u} \sum_{j \geq 0} h_j^{(q-1)} u^j.$$

The family in the numerator is holomorphic at $u = 0$. Comparing the coefficients of $(-1)^n u^n / n!$ proves (4.50). Its Fourier coefficients are $O(|n|^{-q}(1 + \log |n|)^{n_0+1})$ when the fixed Stieltjes index is denoted by n_0 . Therefore the Fourier series and its first $q-2$ ordinary derivatives converge absolutely for $q \geq 2$. The derivative formulas for P_k^θ and Q_k^θ follow directly from (4.25) and $\psi(p) = H_{p-1} - \gamma$.

For (4.53), the case $q = 1, n = 0$ is $-\partial_s Z_s^{1/2}|_{s=0}$. Use (4.33), $\zeta(0, a) = \frac{1}{2} - a$, and $\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$. The difference of the two zeta values at zero is $\frac{1}{2}$, which supplies the term $\frac{1}{2} \log 2$ after differentiation. $\square$

4.7 Degeneration to the untwisted finite parts

The zero Fourier coefficient becomes singular as $\theta \downarrow 0$. Deleting it exactly gives a continuous transition to the untwisted normalization, including all the Stieltjes jets.

Theorem 4.10 (Exact zero-mode renormalization). *For every $\rho < 1$,*

$$B_\theta(u) - \Gamma(1-u)(2\pi i \theta)^u 1 \longrightarrow B(u) \quad (\theta \downarrow 0) \quad (4.54)$$

locally uniformly for $|u| < \rho$ after pairing with any smooth periodic test function, and with every fixed number of derivatives in u . Here 1 is the constant distribution, which has only the zero Fourier coefficient. Put

$$\mu_n(\theta) = \frac{(-1)^{n+1}}{n+1} \mathcal{B}_{n+1}(\gamma + \log(2\pi i \theta), \zeta(2), 2!\zeta(3), \dots, n!\zeta(n+1)). \quad (4.55)$$

Then $\mu_n(\theta) = \widehat{U}_n^\theta(0)$ and

$$U_n^\theta - \mu_n(\theta) 1 \longrightarrow T_n \quad \text{in } \mathcal{D}'(\mathbb{R}/\mathbb{Z}). \quad (4.56)$$

In particular,

$$\mu_0(\theta) = -\gamma - \log(2\pi i \theta), \quad \mu_1(\theta) = \frac{(\gamma + \log(2\pi i \theta))^2 + \zeta(2)}{2}.$$

Proof. The subtracted quantity in (4.54) is exactly the $n = 0$ Fourier coefficient. For each $n \neq 0$, the remaining coefficient tends to $\Gamma(1-u)(2\pi in)^u = \widehat{B(u)}(n)$ with the previously fixed branch. Uniformly for $0 < \theta \leq \frac{1}{2}$ and u on a compact subdisk, these coefficients and their fixed order derivatives are bounded by a common polynomial in $|n|$, because $|n + \theta|$ is comparable with $|n|$. Rapid decrease of the test-function Fourier coefficients proves normal convergence and allows termwise order differentiation. Taking the coefficient of $(-1)^{n+1}u^{n+1}/n!$ gives (4.56); the scalar Gamma expansion gives (4.55). $\square$

At $u = 0$, the left side of (4.54) is already $\delta_0 - 1 = \Delta$. Thus the change of convolution identity is caused exactly by the disappearing zero mode. The same observation explains why the interior alternating identity (4.39) has no added constant, whereas the untwisted digamma convolution has the constant contributed by $-\zeta(2)\Delta$.

5 Shifted harmonic sums, beta derivatives, and exact Laurent data

The all-order Stieltjes differentiation formulas in the manuscript have elementary symmetric harmonic coefficients. Those coefficients can also be put inside a Dirichlet series. This gives a second exact calculus: positive integer values are beta derivatives, whereas all principal parts and finite parts at nonpositive integers are finite polynomials. The dependence on an arbitrary shift is retained throughout.

There are important antecedents. Young's parametric hyperharmonic Euler sums include the beta-derivative evaluation below [18]; his work on height-one multiple zeta functions supplies the unshifted generating-function framework [17], and his earlier paper treats their unshifted Laurent expansions [16]. Harmonic Stieltjes constants at depth one have a substantial separate literature [19]. Accordingly, neither the beta method nor the unshifted height-one theory is asserted to be new. The main continuation developed here is the all-depth coefficient rule for outer-shifted Laurent data at nonpositive integers, together with its polynomial shift calculus and the residue term required when finite parts are differentiated. These formulas were not located in the inspected repository sources; no claim of priority over all literature is made.

5.1 Definitions and a single two-variable family

Put $A = a + 1$ and initially assume $\Re a > -1$. Write

$$e_r(N) = [u^r] \prod_{j=1}^N \left(1 + \frac{u}{j}\right), \quad e_0(N) = 1, \quad e_r(N) = 0 \quad (r > N). \quad (5.1)$$

Thus $e_r(N) = H_N(\{1\}^r)$ is the strict multiple harmonic sum, and

$$\begin{aligned} e_1(N) &= H_N, \\ e_2(N) &= \frac{H_N^2 - H_N^{(2)}}{2}, \\ e_3(N) &= \frac{H_N^3 - 3H_N H_N^{(2)} + 2H_N^{(3)}}{6}. \end{aligned}$$

The main family is

$$E_r(s; a) = \sum_{n=1}^{\infty} \frac{e_r(n-1)}{(n+a)^s}, \quad r \geq 0, \quad \Re s > 1. \quad (5.2)$$

The power uses the principal logarithm; $n+a$ lies in the right half-plane. For $a=0$, this is $\zeta(s, \{1\}^r)$, and for $r=0$ it is $\zeta(s, A)$. The shift affects only the outer denominator. In particular, for general a this is not the harmonic Hurwitz function in which every inner denominator is shifted as well.

The bound $e_r(n-1) \leq H_{n-1}^r/r!$ proves normal convergence of (5.2) on compact subsets of the displayed domain. To keep all depths together, define

$$F(s, u; a) = \sum_{n=1}^{\infty} \frac{\Gamma(n+u)}{\Gamma(n)\Gamma(1+u)} \frac{1}{(n+a)^s}. \quad (5.3)$$

For u near zero and $\Re s > 1 + |u|$, the defining series is normally convergent. The quotient in the numerator is the polynomial $\prod_{j=1}^{n-1} (1 + u/j)$, so

$$F(s, u; a) = \sum_{r \geq 0} E_r(s; a) u^r. \quad (5.4)$$

Equivalently, F is the Barnes–Hurwitz zeta function with complex order $1+u$, in the convention used in [17].

Proposition 5.1 (Mellin representations). *In a common domain of absolute convergence,*

$$\Gamma(s)F(s, u; a) = \int_0^\infty t^{s-1} e^{-At} (1 - e^{-t})^{-1-u} dt, \quad (5.5)$$

$$E_r(s; a) = \frac{1}{r! \Gamma(s)} \int_0^1 \frac{x^a (-\log x)^{s-1} [-\log(1-x)]^r}{1-x} dx. \quad (5.6)$$

For fixed r , the second integral converges for $\Re s > 1$ and $\Re a > -1$.

Proof. The binomial series gives

$$\sum_{n \geq 1} \frac{\Gamma(n+u)}{\Gamma(n)\Gamma(1+u)} x^n = \frac{x}{(1-x)^{1+u}}.$$

Multiply by $x^{a-1}(-\log x)^{s-1}$ and integrate. Absolute convergence permits termwise integration, and the integral of the n th term is $\Gamma(s)(n+a)^{-s}$. The substitution $x = e^{-t}$ gives (5.5); coefficient extraction in u gives (5.6). At $x=1$, its absolute integrand is bounded by a constant times $(1-x)^{\Re s-2}(1+|\log(1-x)|^r)$, which is integrable for $\Re s > 1$. At zero, $\Re a > -1$ suffices. $\square$

This also gives a direct polylogarithmic interpretation, since

$$\text{Li}_{\{1\}^{r+1}}(x) = \frac{[-\log(1-x)]^{r+1}}{(r+1)!}, \quad \frac{d}{dx} \text{Li}_{\{1\}^{r+1}}(x) = \frac{[-\log(1-x)]^r}{r!(1-x)}.$$

Thus (5.6) is the Mellin transform of a derivative of a multiple polylogarithm of arbitrary depth.

5.2 Positive integer values and their antiderivatives

Define generalized harmonic functions by

$$H_a^{(1)} = \psi(A) + \gamma, \quad H_a^{(k)} = \zeta(k) - \zeta(k, A) \quad (k \geq 2). \quad (5.7)$$

They agree with the usual finite harmonic numbers at nonnegative integer a . Introduce

$$R_a(u) = \frac{\Gamma(1-u)\Gamma(A)}{\Gamma(A-u)} = \exp\left(\sum_{k \geq 1} \frac{H_a^{(k)}}{k} u^k\right) = \sum_{d \geq 0} h_d(a) u^d. \quad (5.8)$$

Here $h_d(a)$ is a complete symmetric harmonic polynomial:

$$h_0(a) = 1, \quad d h_d(a) = \sum_{k=1}^d H_a^{(k)} h_{d-k}(a). \quad (5.9)$$

For example,

$$h_1 = H_a^{(1)}, \quad h_2 = \frac{(H_a^{(1)})^2 + H_a^{(2)}}{2}, \quad h_3 = \frac{(H_a^{(1)})^3 + 3H_a^{(1)}H_a^{(2)} + 2H_a^{(3)}}{6}.$$

The plus signs distinguish these polynomials from the elementary coefficients $e_r(N)$ in the summand.

Theorem 5.2 (Subtracted beta identity and all positive integer values). *For $r \geq 0$ and $\Re a > -1$,*

$$\boxed{h_{r+1}(a) = \sum_{n \geq 1} e_r(n-1) \left(\frac{1}{n} - \frac{1}{n+a} \right)}. \quad (5.10)$$

For every integer $p \geq 2$,

$$\boxed{E_r(p; a) = \frac{(-1)^p}{(p-1)!} \frac{d^{p-1}}{da^{p-1}} h_{r+1}(a)}. \quad (5.11)$$

Equivalently, as an analytic germ at $(x, u) = (0, 0)$,

$$\sum_{p \geq 2} \sum_{r \geq 0} E_r(p; a) x^{p-1} u^r = B(A-x, -u) - B(A, -u). \quad (5.12)$$

The poles of the individual beta factors at $u = 0$ cancel in this difference.

Proof. Initially take $\Re u < 0$. The beta integral and (5.5) at $s = 1$ give

$$F(1, u; a) = B(A, -u) = -\frac{R_a(u)}{u}, \quad F(1, u; 0) = -\frac{1}{u}.$$

The difference of their series is normally convergent also for u near zero: its n th term is $O(n^{|u|-2})$ on compact a -sets. Consequently

$$\sum_{n \geq 1} \prod_{j < n} (1 + u/j) \left(\frac{1}{n} - \frac{1}{n+a} \right) = \frac{R_a(u) - 1}{u}$$

extends holomorphically through $u = 0$. Compare coefficients to obtain (5.10). Normal convergence permits $p - 1$ derivatives in a , giving (5.11).

For an independent verification of the bivariate formula, sum (5.6) first over $p \geq 2$ and then over r . The result is the convergent integral

$$\int_0^1 x_0^{A-1} (x_0^{-x} - 1) (1 - x_0)^{-1-u} dx_0.$$

For $|x|$ sufficiently small and $|u| < 1/2$, it is integrable at both endpoints. In the subdomain $\Re u < 0$ it is a difference of two beta integrals. Analytic continuation across $u = 0$ proves (5.12) without separating divergent integrals there. Finally, Taylor expansion of $\log \Gamma$ proves (5.8), and logarithmic differentiation gives (5.9). $\square$

Formula (5.11) is a specialization and explicit coefficient implementation of the established parametric beta theory [18]. Its useful feature for this manuscript is that it simultaneously evaluates infinitely many nonlinear harmonic combinations and gives their antiderivatives. In particular, $\int E_r(2; a) da = h_{r+1}(a) + C$.

Put $Z_k = \zeta(k, A)$ and $H = H_a^{(1)}$. Since $\partial_a H_a^{(k)} = k Z_{k+1}$, the first outer exponent has the especially short all-depth expression

$$E_r(2; a) = \sum_{k=1}^{r+1} Z_{k+1} h_{r+1-k}(a). \quad (5.13)$$

For example,

$$E_1(2; a) = H Z_2 + Z_3, \quad (5.14)$$

$$E_2(2; a) = h_2(a) Z_2 + H Z_3 + Z_4, \quad (5.15)$$

$$E_1(3; a) = H Z_3 + \frac{3}{2} Z_4 - \frac{1}{2} Z_2^2, \quad (5.16)$$

$$E_2(3; a) = h_2(a) Z_3 + \frac{3}{2} H Z_4 + 2 Z_5 - \frac{1}{2} H Z_2^2 - Z_2 Z_3. \quad (5.17)$$

At $a = 0$, (5.13) recovers $\zeta(2, \{1\}^r) = \zeta(r+2)$; the next two examples give $\zeta(3, 1) = \zeta(4)/4$ and $\zeta(3, 1, 1) = 2\zeta(5) - \zeta(2)\zeta(3)$.

5.3 An all-depth cyclotomic projection

Use the convention

$$\text{Li}_{p, \{1\}^r}(z_0, z_1, \dots, z_r) = \sum_{m_0 > m_1 > \dots > m_r \geq 1} \frac{z_0^{m_0} z_1^{m_1} \dots z_r^{m_r}}{m_0^p m_1 \dots m_r}.$$

For $p \geq 2$ this series is absolutely convergent when all the z_j are roots of unity.

Corollary 5.3 (Explicit reduction of a cyclotomic average). *Let $q \geq 1$, $1 \leq h \leq q$, $\omega = e^{2\pi i/q}$, and $a = h/q - 1$. For every $p \geq 2$ and $r \geq 0$,*

$$\sum_{j_0, \dots, j_r=0}^{q-1} \omega^{-h j_0} \text{Li}_{p, \{1\}^r}(\omega^{j_0}, \dots, \omega^{j_r}) = q^{1-p} \frac{(-1)^p}{(p-1)!} h_{r+1}^{(p-1)}(h/q - 1). \quad (5.18)$$

Every value on the right reduces to depth-one polylogarithms at q th roots of unity and logarithms. This is a reduction of the specified average; it makes no assertion that its individual colored summands all reduce to depth one.

Proof. In (5.2), set $m_0 = q(n-1) + h$ and $m_i = qk_i$ for the indices $n-1 \geq k_1 > \dots > k_r \geq 1$ in $e_r(n-1)$. This gives

$$E_r(p; h/q - 1) = q^{p+r} \sum_{\substack{m_0 > \dots > m_r \geq 1 \\ m_0 \equiv h \pmod{q}, m_i \equiv 0 \pmod{q}, i \geq 1}} \frac{1}{m_0^p m_1 \dots m_r}.$$

Insert the finite Fourier filters for the $r+1$ congruences and apply (5.11). To make the depth-one reduction explicit, one has, for $k \geq 2$,

$$\zeta(k, h/q) = q^{k-1} \sum_{j=0}^{q-1} \omega^{-hj} \text{Li}_k(\omega^j), \quad (5.19)$$

$$H_{h/q-1}^{(1)} = -\log q + \sum_{j=1}^{q-1} \omega^{-hj} \log(1 - \omega^j). \quad (5.20)$$

The first identity follows by grouping a Dirichlet series into residue classes. Expand the same identity at spectral argument 1 to obtain the second, using $\text{Li}_1(z) = -\log(1-z)$ with principal logarithms. Complex conjugate terms have conjugate logarithms, so (5.20) is real. Formula (5.7) and the recurrence (5.9) finish the reduction. $\square$

For a concrete nonlinear family, let $L = \log 2$. Specializing to $a = -1/2$ gives

$$\sum_{n \geq 0} \frac{H_n}{(2n+1)^2} = \frac{7\zeta(3) - \pi^2 L}{4}, \quad (5.21)$$

$$\sum_{n \geq 0} \frac{H_n^2 - H_n^{(2)}}{(2n+1)^2} = \frac{\pi^2 L^2}{2} - 7L\zeta(3) + \frac{\pi^4}{24}, \quad (5.22)$$

$$\begin{aligned} \sum_{n \geq 0} \frac{H_n^3 - 3H_n H_n^{(2)} + 2H_n^{(3)}}{(2n+1)^2} &= \frac{3}{2} [31\zeta(5) - 13\zeta(2)\zeta(3) - 15L\zeta(4) \\ &\quad + 14L^2\zeta(3) - 4L^3\zeta(2)]. \end{aligned} \quad (5.23)$$

All three are instances of one identity, valid as a germ at $u = 0$:

$$\sum_{r \geq 0} u^r \sum_{n \geq 0} \frac{e_r(n)}{(2n+1)^2} = \frac{\sqrt{\pi} \Gamma(1-u)}{4u \Gamma(1/2-u)} [\psi(1/2) - \psi(1/2-u)]. \quad (5.24)$$

This yields arbitrary degree in the harmonic numbers, with no integer-relation search. Assigning weight k to $H_a^{(k)}$ and $\zeta(k, A)$ makes (5.11) homogeneous of weight $p+r$; the rational Fourier reduction preserves this weight.

5.4 Meromorphic continuation with a uniform remainder

The same family gives exact spectral data beyond the half-plane where its Dirichlet series converges. Define polynomials $b_k(u, a)$ by

$$\phi(t, u, a) := e^{-At} \left(\frac{t}{1 - e^{-t}} \right)^{1+u} = \sum_{k \geq 0} b_k(u, a) t^k. \quad (5.25)$$

The branch at $t = 0$ is the analytic one taking value 1. Equivalently, in the standard generalized Bernoulli-polynomial notation,

$$b_k(u, a) = \frac{B_k^{(1+u)}(u - a)}{k!}.$$

It is enough to use the defining series; no properties of generalized Bernoulli polynomials are assumed. In particular,

$$b_0 = 1, \quad (5.26)$$

$$b_1 = \frac{u - 2a - 1}{2}, \quad (5.27)$$

$$b_2 = \frac{12a^2 - 12au + 12a + 3u^2 - 7u + 2}{24}, \quad (5.28)$$

$$b_3 = \frac{(u - 2a - 1)(4a^2 - 4au + 4a + u^2 - 3u)}{48}. \quad (5.29)$$

Each b_k is a rational polynomial in a, u , of degree at most k in each variable.

Lemma 5.4 (Uniform Mellin subtraction). *Fix a compact subset of $\Re a > -1$ and a sufficiently small disc of u -values around zero. For every integer $K \geq 1$,*

$$\Gamma(s)F(s, u; a) = \sum_{k=0}^{K-1} \frac{b_k(u, a)}{s - 1 - u + k} + J_K(s, u; a), \quad (5.30)$$

where J_K is jointly holomorphic on $\Re s > 1 + \Re u - K$. Therefore F extends meromorphically to all s , locally in u , and $E_r(s; a) = [u^r]F(s, u; a)$ has possible poles only at $s = 1, 0, -1, -2, \dots$

Proof. Split (5.5) at $t = 1$, and define

$$\begin{aligned} J_K(s, u; a) &= \int_0^1 t^{s-2-u} \left(\phi(t, u, a) - \sum_{k=0}^{K-1} b_k(u, a) t^k \right) dt \\ &\quad + \int_1^\infty t^{s-1} e^{-At} (1 - e^{-t})^{-1-u} dt. \end{aligned}$$

The parenthesized remainder is $O(t^K)$ uniformly on the fixed parameter sets. Every derivative in s or u introduces only a fixed power of $\log t$ near zero, so it remains integrable on compact subsets of $\Re s > 1 + \Re u - K$. The second integral and all its parameter derivatives decay exponentially at infinity because $\Re A$ stays positive. Differentiation under both integrals proves joint holomorphy. Integrating each subtracted monomial gives (5.30). Increasing K continues the same function, since all formulas agree on the original convergence domain. Finally, $1/\Gamma(s)$ is entire; coefficient extraction in u can turn the simple moving denominators into higher poles at the stated integers, and introduces no others. $\square$

5.5 The complete principal part and finite part at $s = 1$

Write

$$C(u) = \frac{1}{\Gamma(1+u)} = \sum_{j \geq 0} c_j u^j = \exp \left(\gamma u + \sum_{k \geq 2} \frac{(-1)^{k+1} \zeta(k)}{k} u^k \right). \quad (5.31)$$

Thus $c_0 = 1$, $c_1 = \gamma$, $c_2 = (\gamma^2 - \zeta(2))/2$, and $c_3 = (\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3))/6$. The finite part $\text{FP}_{s=s_0}$ always means the coefficient of $(s - s_0)^0$ in the Laurent expansion of the specified meromorphic function.

Theorem 5.5 (All depths at the principal pole). *For $r \geq 0$ and $\Re a > -1$, as $\varepsilon \rightarrow 0$,*

$$E_r(1 + \varepsilon; a) = \sum_{j=0}^r \frac{c_j}{\varepsilon^{r+1-j}} + c_{r+1} - h_{r+1}(a) + O(\varepsilon). \quad (5.32)$$

In particular, all principal coefficients are independent of the shift, while the complete shift dependence of the finite part is the single complete harmonic polynomial $-h_{r+1}(a)$.

Proof. First take $a = 0$. With $K = 1$, the singular term in (5.30) near $(s, u) = (1, 0)$ is $1/(s - 1 - u)$. Consequently

$$F(s, u; 0) = \frac{1}{\Gamma(1+u)(s-1-u)} + Q(s, u),$$

with Q jointly holomorphic there: replacing $1/\Gamma(s)$ by $1/\Gamma(1+u)$ in the singular numerator changes the expression by a holomorphic divided difference. For $s = 1$ and small $\Re u < 0$, the beta integral gives $F(1, u; 0) = -1/u$, and hence

$$Q(1, u) = \frac{C(u) - 1}{u}.$$

The right side extends holomorphically to $u = 0$. Extracting $[u^r]$ from $C(u)/(\varepsilon - u)$ gives exactly $\sum_{j=0}^r c_j \varepsilon^{-r-1+j}$, and extracting it from $Q(1, u)$ gives c_{r+1} .

For arbitrary a , the difference

$$E_r(s; a) - E_r(s; 0) = \sum_{n \geq 1} e_r(n-1) ((n+a)^{-s} - n^{-s})$$

is normally convergent for $\Re s > 0$, locally uniformly in a . At $s = 1$ it equals $-h_{r+1}(a)$ by (5.10). This leaves the principal part unchanged and gives the stated finite part. $\square$

For example,

$$\text{FP}_{s=1} E_0(s; a) = -\psi(a), \quad (5.33)$$

$$\text{FP}_{s=1} E_1(s; a) = \frac{\gamma^2 - \zeta(2) - (H_a^{(1)})^2 - H_a^{(2)}}{2}, \quad (5.34)$$

$$\text{FP}_{s=1} E_2(s; a) = \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6} - h_3(a). \quad (5.35)$$

The first line is the usual zeroth generalized Stieltjes constant. For higher depths, the corresponding zeroth constants are just as explicit. This does not imply such a reduction for all higher Laurent coefficients.

5.6 Every nonpositive integer: a finite exact coefficient rule

Theorem 5.6 (Principal and finite Laurent data at nonpositive integers). *Let $m, r \geq 0$, and define*

$$\mathcal{A}_m(u, a) = \frac{b_{m+1}(u, a)}{\Gamma(u - m)} = \sum_{d \geq 1} A_{m,d}(a) u^d. \quad (5.36)$$

Then

$$E_r(-m + \varepsilon; a) = \sum_{h=0}^r A_{m,r+1-h}(a) \varepsilon^{-h} + O(\varepsilon). \quad (5.37)$$

In particular,

$$\text{FP}_{s=-m} E_r(s; a) = [u^{r+1}] \frac{b_{m+1}(u, a)}{\Gamma(u - m)}. \quad (5.38)$$

All coefficients displayed in (5.37) belong to $\mathbb{Q}[a, \gamma, \zeta(2), \dots, \zeta(r)]$, with the obvious interpretation for $r = 0, 1$. The highest possible pole coefficient is

$$A_{m,1}(a) = \zeta(-m, A). \quad (5.39)$$

Proof. Choose $K > m + 1$ in Lemma 5.4. Near $(s, u) = (-m, 0)$, precisely one of its denominators can vanish: the one with $k = m + 1$. Thus, putting $s = -m + \varepsilon$,

$$F(-m + \varepsilon, u; a) = \frac{b_{m+1}(u, a)}{\Gamma(-m + \varepsilon)(\varepsilon - u)} + \frac{Q_m(\varepsilon, u; a)}{\Gamma(-m + \varepsilon)},$$

where Q_m is jointly holomorphic. Since $1/\Gamma(-m + \varepsilon)$ has a simple zero at $\varepsilon = 0$, the second term contributes $O(\varepsilon)$ after extracting any fixed coefficient in u . Write

$$b_{m+1}(u, a) = \sum_{j \geq 0} b_{m+1,j}(a) u^j, \quad \frac{1}{\Gamma(-m + \varepsilon)} = \sum_{\ell \geq 1} g_{m,\ell} \varepsilon^\ell.$$

Coefficient extraction from the singular term gives

$$\left(\sum_{\ell \geq 1} g_{m,\ell} \varepsilon^\ell \right) \sum_{j=0}^r b_{m+1,j}(a) \varepsilon^{-r+j-1}.$$

Its coefficient at ε^{-h} is $\sum_{j=0}^{r-h} b_{m+1,j}(a) g_{m,r+1-h-j}$, which is exactly $[u^{r+1-h}] \mathcal{A}_m(u, a)$. This proves (5.37).

The reciprocal-gamma factor has the explicit harmonic expansion

$$\frac{1}{\Gamma(u - m)} = (-1)^m m! u \exp \left((\gamma - H_m) u + \sum_{k \geq 2} \frac{(-1)^{k+1} \zeta(k) - H_m^{(k)}}{k} u^k \right). \quad (5.40)$$

Indeed, multiply $1/\Gamma(1 + u)$ by $u \prod_{j=1}^m (u - j)$ and expand its logarithm after removing $(-1)^m m! u$. The finite coefficient rule now proves the claimed polynomial ring. Finally, $b_{m+1}(0, a) = B_{m+1}(-a)/(m+1)!$ and $g_{m,1} = (-1)^m m!$. Bernoulli reflection and $\zeta(-m, A) = -B_{m+1}(A)/(m+1)$ give (5.39). $\square$

For a generic shift, the pole at $s = -m$ has order r when $r \geq 1$. At a zero of $\zeta(-m, A)$ its order drops. Thus the familiar absence of negative even poles for the depth-one unshifted harmonic zeta function is part of an all-depth cancellation rule: at $a = 0$ and positive even m , the pole order is at most $r - 1$. The formula determines whether further cancellations occur.

No divergent series is assigned an ordinary sum here. The values in (5.38) are finite Laurent coefficients of a specified meromorphic continuation. Nor can they be computed by first setting $s = -m$ in F at a nonzero u and then differentiating: the moving divisor $\varepsilon - u = 0$ makes those operations incompatible at their intersection.

5.7 Small exact tables and differentiation of finite parts

At $s = 0$, (5.27) makes the whole finite-part family particularly simple. With $c_{-1} = 0$,

$$\text{FP}_{s=0} E_r(s; a) = \frac{1}{2} c_{r-1} - \left(a + \frac{1}{2}\right) c_r. \quad (5.41)$$

More generally every finite part at $s = -m$ is a polynomial in a of degree at most $m + 1$. The first unshifted examples are shown in Table 1; only finite parts, not complete Laurent series, are tabulated.

	$s = 0$	$s = -1$	$s = -2$
$r = 0$	$-\frac{1}{2}$	$-\frac{1}{12}$	0
$r = 1$	$\frac{1-\gamma}{2}$	$\frac{9-2\gamma}{24}$	$\frac{1}{8}$
$r = 2$	$\frac{-\gamma^2+2\gamma+\zeta(2)}{4}$	$\frac{-\gamma^2+9\gamma+\zeta(2)-10}{24}$	$\frac{6\gamma-17}{48}$

Table 1: Exact finite parts of $E_r(s; 0)$ from the single coefficient rule (5.38).

For example, the complete displayed portion at a point where the highest possible pole cancels is

$$E_3(-2 + \varepsilon; 0) = \frac{1}{8\varepsilon^2} + \frac{6\gamma - 17}{48\varepsilon} + \frac{3\gamma^2 - 17\gamma - 3\zeta(2) + 17}{48} + O(\varepsilon). \quad (5.42)$$

There is a necessary residue correction when finite parts are differentiated. Denote

$$C_{m,r}(a) = \text{FP}_{s=-m} E_r(s; a) \quad (m \geq 0), \quad C_{-1,r}(a) = \text{FP}_{s=-1} E_r(s; a),$$

and let $R_{m,r}(a)$ denote the residue at $s = -m$, also for $m = -1$.

Corollary 5.7 (Differentiation with its residue term). *For $m \geq 0$,*

$$C'_{m,r}(a) = m C_{m-1,r}(a) - R_{m-1,r}(a). \quad (5.43)$$

In particular $C'_{0,r}(a) = -c_r$.

Proof. Normal convergence first gives $\partial_a E_r(s; a) = -s E_r(s + 1; a)$ for $\Re s > 1$; Lemma 5.4 continues the identity meromorphically. Put $s = -m + \varepsilon$. The constant term of $(m - \varepsilon) E_r(1 - m + \varepsilon; a)$ is $m C_{m-1,r} - R_{m-1,r}$, even if the latter function has higher order poles. Taking the constant coefficient commutes with a -differentiation by joint meromorphy on compact shift domains. At $m = 0$, Theorem 5.5 gives $R_{-1,r} = c_r$, in agreement with (5.41). $\square$

There is also a direct coefficient-level verification. Since $\partial_a b_{m+1}(u, a) = -b_m(u, a)$ and $\Gamma(u - m + 1) = (u - m)\Gamma(u - m)$,

$$\partial_a \mathcal{A}_m(u, a) = (m - u)\mathcal{A}_{m-1}(u, a), \quad m \geq 1.$$

Taking the coefficient of u^{r+1} gives exactly (5.43); the case $m = 0$ is (5.41).

Dropping the residue in (5.43) would, already at $m = 0$, incorrectly predict a constant function of a . This is a correction to a tempting manipulation, rather than an assertion that the inspected manuscript makes that error.

5.8 Higher Stieltjes coefficients and the precise scope of reduction

Define the higher shifted height-one constants by

$$E_r(1 + \varepsilon; a) = \sum_{j=0}^r c_j \varepsilon^{-r-1+j} + \sum_{\ell \geq 0} \frac{(-1)^\ell}{\ell!} \gamma_{r,\ell}(a) \varepsilon^\ell. \quad (5.44)$$

For $r = 0$, these are the usual generalized Stieltjes constants $\gamma_{0,\ell}(a) = \gamma_\ell(A)$. Theorem 5.5 evaluates $\gamma_{r,0}(a) = c_{r+1} - h_{r+1}(a)$. For real $a, b > -1$ and every $\ell \geq 0$, normal convergence of the shifted difference gives the exact, absolutely convergent identity

$$\gamma_{r,\ell}(a) - \gamma_{r,\ell}(b) = \sum_{n \geq 1} e_r(n-1) \left(\frac{\log^\ell(n+a)}{n+a} - \frac{\log^\ell(n+b)}{n+b} \right). \quad (5.45)$$

The summand is $O(n^{-2}(1 + \log^{r+\ell} n))$. At $\ell = 0$ this reduces to $h_{r+1}(b) - h_{r+1}(a)$. At higher ℓ it specifies the shift dependence without assuming a finite reduction of the individual constants.

One also obtains the all-depth extension of the manuscript's Stieltjes differentiation ladder. Let $e_i(k) = [v^i] \prod_{j=1}^k (1 + v/j)$ and use ∂_s for spectral differentiation. For $k \geq 1$,

$$\frac{d^k}{da^k} \gamma_{r,\ell}(a) = (-1)^{\ell+k} k! \ell! \sum_{i+j=\ell} e_i(k) \frac{\partial_s^j E_r(k+1; a)}{j!}. \quad (5.46)$$

Indeed, $\partial_a^k E_r(s; a) = (-1)^k (s)_k E_r(s+k; a)$; the principal part at $s = 1$ is independent of a , and coefficient comparison at $s = 1$ proves the formula. When $\ell = 0$ the right side is fully evaluated by Theorem 5.2. For $\ell > 0$ it involves genuine spectral derivatives.

For completeness these derivatives have exact logarithmic Mellin representations. Differentiating (5.6) yields

$$\begin{aligned} \frac{1}{r!} \int_0^1 \frac{x^a (-\log x)^{s-1} [-\log(1-x)]^r}{1-x} \log^\ell(-\log x) dx \\ = \sum_{j=0}^{\ell} \binom{\ell}{j} \Gamma^{(\ell-j)}(s) \partial_s^j E_r(s; a), \quad \Re s > 1. \end{aligned} \quad (5.47)$$

The endpoint estimates in Proposition 5.1 remain integrable after any fixed number of logarithmic factors, justifying all differentiations. Gamma derivatives here can equivalently be written using complete Bell polynomials in polygamma functions.

The exact polynomial conclusion of Theorem 5.6 stops at the constant Laurent coefficient. The term $Q_m/\Gamma(s)$ in its proof starts at order $s + m$ and can contribute new constants from that point onward. At depth zero these already include $\zeta'(-m, A)$, and at $m = 0$ Lerch's identity gives the log-gamma function. There is therefore no basis for discarding the analytic remainder when computing positive spectral derivatives.

The next concrete questions are to identify which linear combinations of these higher coefficients eliminate the analytic remainder, to extend the cyclotomic projection to harmonic patterns other than $\{1\}^r$, and to compare the outer-shift constants (5.44) systematically with the literature's simultaneously shifted harmonic Hurwitz constants. Those extensions are left open here.

5.9 Reflection and antiderivatives of the harmonic finite parts

The shifted coefficient theorem also gives exact transformation laws that are not visible in an unshifted table. Put

$$C_{m,r}(a) = \text{FP}_{s=-m} E_r(s; a) = [u^{r+1}] \mathcal{A}_m(u, a), \quad m, r \geq 0,$$

and set $C_{m,r} = 0$ for $r < 0$. Each $C_{m,r}$ is a polynomial in a , of degree at most $m + 1$. Thus the identities below hold as polynomial identities for all complex a ; they refer directly to the original shifted Dirichlet family where $\Re a > -1$.

Theorem 5.8 (Reflection mixes harmonic depths). *For all $m, r \geq 0$,*

$$C_{m,r}(-1 - a) = (-1)^{m+1} \sum_{j=0}^r \frac{1}{j!} \frac{d^j}{da^j} C_{m,r-j}(a). \quad (5.48)$$

Terms with $j > m + 1$ vanish. On the strip $-1 < \Re a < 0$, both sides are finite parts of the original shifted families. Elsewhere the left side means the polynomial continuation of its finite-part coefficient.

Proof. The defining local kernel can be written

$$\phi(t, u, a) = e^{((u-1)/2-a)t} \left(\frac{t}{2 \sinh(t/2)} \right)^{1+u}.$$

All powers here mean their Taylor germs at $t = 0$, where the parenthesized factor equals one. Its second factor is even in t . Consequently

$$\phi(t, u, -1 - a) = \phi(-t, u, a + u), \quad b_{m+1}(u, -1 - a) = (-1)^{m+1} b_{m+1}(u, a + u).$$

The Gamma denominator is independent of a , so the same identity holds for $\mathcal{A}_m$. Expanding its polynomial argument translation in powers of u and taking $[u^{r+1}]$ proves (5.48). There is no constant coefficient of $\mathcal{A}_m$, which is why the sum ends at $j = r$. $\square$

For $r = 0$, this is the familiar Bernoulli reflection $\zeta(-m, -a) = (-1)^{m+1} \zeta(-m, a + 1)$, interpreted as a polynomial identity. At positive depth the derivative terms are essential. For example,

$$C_{1,1}(-1 - a) - C_{1,1}(a) = C'_{1,0}(a) = -a - \frac{1}{2}.$$

Thus an ordinary Bernoulli parity rule at fixed depth would be false.

Theorem 5.9 (Complete finite-part antiderivative ladder). *Let $q \geq 1$, and define $h_j^{[m,q]}$ by*

$$\prod_{v=1}^q \left(1 - \frac{u}{m+v}\right)^{-1} = \sum_{j \geq 0} h_j^{[m,q]} u^j. \quad (5.49)$$

Then the polynomial

$$F_{m,r}^{(q)}(a) = \frac{1}{(m+1)_q} \sum_{j=0}^r h_j^{[m,q]} C_{m+q,r-j}(a) \quad (5.50)$$

satisfies

$$\frac{d^q}{da^q} F_{m,r}^{(q)}(a) = C_{m,r}(a).$$

Here $(m+1)_q = (m+1)(m+2) \cdots (m+q)$. Every q -fold antiderivative differs from this one by a polynomial of degree less than q .

Proof. Differentiating the local kernel gives $\partial_a b_k = -b_{k-1}$, hence

$$\partial_a \mathcal{A}_m = (m-u) \mathcal{A}_{m-1}.$$

Iteration, in the direction of increasing m , gives

$$\partial_a^q \mathcal{A}_{m+q} = \prod_{v=1}^q (m+v-u) \mathcal{A}_m.$$

Divide by this polynomial, which is invertible as a germ at $u = 0$, and extract $[u^{r+1}]$. Equation (5.49) gives exactly (5.50). $\square$

The coefficients in this ladder are again generalized-harmonic Bell coefficients:

$$h_j^{[m,q]} = [u^j] \exp \left(\sum_{k \geq 1} \frac{H_{m+q}^{(k)} - H_m^{(k)}}{k} u^k \right).$$

In particular,

$$\int C_{m,r}(a) da = \sum_{j=0}^r \frac{C_{m+1,r-j}(a)}{(m+1)^{j+1}} + \text{constant}. \quad (5.51)$$

The lower-depth terms are exactly the residue correction propagated through integration. To normalize a repeated integral at a_0 , use

$$F_{m,r}^{(q)}(a) - \sum_{j=0}^{q-1} \frac{(a-a_0)^j}{j!} \frac{d^j}{da^j} F_{m,r}^{(q)}(a) \Big|_{a=a_0}.$$

This polynomial and its first $q-1$ derivatives vanish at a_0 ; it is the q -fold integral with those initial conditions.

5.10 Rational multiplication with explicit depth corrections

An ordinary Hurwitz multiplication identity does not survive unchanged when the outer denominator is shifted but the inner harmonic denominators are fixed. Its finite-part replacement is nevertheless a finite exact formula.

For an integer $q \geq 2$, let

$$\begin{aligned}\Lambda_q(t) &= \log \left(\frac{1 - e^{-t}}{q(1 - e^{-t/q})} \right), \\ d_n^{(q)}(u) &= [t^n] \exp(-u\Lambda_q(t)).\end{aligned}\tag{5.52}$$

The logarithm is its analytic germ with value zero at $t = 0$. Each $d_n^{(q)}$ is a rational polynomial in u of degree at most n , and $d_n^{(q)}(0) = 0$ for $n > 0$. Set

$$\mathcal{A}_{-1}(u, a) = \frac{1}{\Gamma(1+u)} = \sum_{j \geq 0} c_j u^j.$$

This auxiliary term is independent of a ; it is not the generating function of the finite parts at $s = 1$.

Theorem 5.10 (Multiplication law for the singular coefficient germ). *For $m \geq 0$, put $a_j = (a + 1 + j)/q - 1$. Then*

$$\sum_{j=0}^{q-1} \mathcal{A}_m(u, a_j) = \sum_{n=0}^{m+1} q^{n-m} d_n^{(q)}(u) (u-m)_n \mathcal{A}_{m-n}(u, a).\tag{5.53}$$

All factors are analytic germs at $u = 0$. Taking $[u^{r+1}]$ gives a finite multiplication formula for $C_{m,r}$; taking lower powers gives the corresponding principal Laurent coefficients.

Proof. Sum the defining local kernels at the indicated shifts:

$$\begin{aligned}\sum_{j=0}^{q-1} \phi(t, u, a_j) &= e^{-(a+1)t/q} \frac{1 - e^{-t}}{1 - e^{-t/q}} \left(\frac{t}{1 - e^{-t}} \right)^{1+u} \\ &= q \phi(t/q, u, a) \exp(-u\Lambda_q(t)).\end{aligned}$$

Taking $[t^{m+1}]$ and dividing by $\Gamma(u-m)$ gives

$$\sum_{j=0}^{q-1} \mathcal{A}_m(u, a_j) = \sum_{n=0}^{m+1} q^{n-m} d_n^{(q)}(u) \frac{b_{m+1-n}(u, a)}{\Gamma(u-m)}.$$

The Gamma recurrence identifies the final quotient as $(u-m)_n \mathcal{A}_{m-n}$, including $n = m+1$, since $b_0 = 1$. This proves the identity without interchanging a singular specialization with an infinite sum. $\square$

At depth zero the correction terms vanish, and the formula reduces to

$$\sum_{j=0}^{q-1} \zeta \left(-m, \frac{a+1+j}{q} \right) = q^{-m} \zeta(-m, a+1).$$

For positive depth, the finite correction terms measure the failure of that unmodified rule.

The first instance is especially simple:

$$\sum_{j=0}^{q-1} C_{0,r}(a_j) = C_{0,r}(a) + \frac{q-1}{2} c_{r-1}, \quad c_j = 0 \quad (j < 0). \quad (5.54)$$

Indeed $d_1^{(q)}(u) = u(q-1)/(2q)$. For $q = 2, m = 1$,

$$d_1^{(2)}(u) = \frac{u}{4}, \quad d_2^{(2)}(u) = \frac{u^2 - u}{32},$$

so

$$\mathcal{A}_1\left(u, \frac{a-1}{2}\right) + \mathcal{A}_1\left(u, \frac{a}{2}\right) = \frac{1}{2} \mathcal{A}_1(u, a) + \frac{u(u-1)}{4} \mathcal{A}_0(u, a) + \frac{u^2(u-1)^2}{16\Gamma(1+u)}. \quad (5.55)$$

For example, its depth-one constant term gives

$$C_{1,1}\left(\frac{a-1}{2}\right) + C_{1,1}\left(\frac{a}{2}\right) = \frac{1}{2} C_{1,1}(a) + \frac{a}{4} + \frac{3}{16}.$$

5.11 Half-shift Laurent identities

The coefficient rule and its duplication law give concise values at the half shift. With $a = -1/2$,

$$b_2(u, -1/2) = \frac{3u^2 - u - 1}{24}, \quad b_3(u, -1/2) = \frac{u(u^2 - u - 1)}{48}.$$

Combining these with $1/\Gamma(u-m)$ yields

$$E_1(-1 + \varepsilon; -1/2) = \frac{1}{24\varepsilon} + \frac{\gamma}{24} + O(\varepsilon), \quad (5.56)$$

$$E_2(-1 + \varepsilon; -1/2) = \frac{1}{24\varepsilon^2} + \frac{\gamma}{24\varepsilon} + \frac{\gamma^2 - \zeta(2) - 8}{48} + O(\varepsilon), \quad (5.57)$$

$$E_1(-2 + \varepsilon; -1/2) = -\frac{1}{24} + O(\varepsilon), \quad (5.58)$$

$$E_2(-2 + \varepsilon; -1/2) = -\frac{1}{24\varepsilon} + \frac{1 - 2\gamma}{48} + O(\varepsilon). \quad (5.59)$$

For example, $E_1(s; -1/2) = \sum_{n \geq 1} H_{n-1}(n-1/2)^{-s}$ in its convergent half-plane. The displayed values mean its meromorphic continuation, not convergent sums at negative s . The drop in pole order at -2 follows from the vanishing $\zeta(-2, 1/2) = 0$; its next coefficient does not vanish. These formulas illustrate exact cancellation across depth, shift, and spectral order.

6 An explicit polylogarithm realization of the convolution calculus

The circle Fourier description gives a direct bridge to the polylogarithm's order derivatives at nonpositive integers. The underlying Fourier summation is classical [10, 8]. The full mixed kernel and the symmetric first-Stieltjes specialization also occur in the archived continuations [3, 4]. This section consolidates their order-derivative form while preserving the endpoint distributions.

6.1 A two-parameter kernel

For $\alpha, \beta \in \mathbb{C}$, define the zero-mean distribution $\mathcal{N}_{\alpha, \beta}$ by

$$\widehat{\mathcal{N}_{\alpha, \beta}}(n) = (2\pi i n)^\alpha (-2\pi i n)^\beta \quad (n \neq 0), \quad \widehat{\mathcal{N}_{\alpha, \beta}}(0) = 0. \quad (6.1)$$

The two powers use the logarithms specified in Section 1. On every compact set of parameters their coefficients and any fixed number of parameter derivatives have a common polynomial growth bound. This proves that $\mathcal{N}$ is entire as a distribution-valued function.

Theorem 6.1 (Polylogarithm kernel for every mixed order). *For $0 < h < 1$, put $z = e^{2\pi i h}$. Then*

$$\begin{aligned} \mathcal{N}_{\alpha, \beta}(h) = (2\pi)^{\alpha+\beta} & \left[e^{\pi i(\alpha-\beta)/2} \text{Li}_{-\alpha-\beta}(z) \right. \\ & \left. + e^{-\pi i(\alpha-\beta)/2} \text{Li}_{-\alpha-\beta}(z^{-1}) \right]. \end{aligned} \quad (6.2)$$

The formula represents the restriction to the open interval of the entire distribution family, not an assertion of ordinary convergence of its Fourier series at every order.

Let P_k be the spectral finite-part polygamma distribution, so that

$$\widehat{P}_k(n) = (2\pi i n)^k [\gamma + \log(2\pi i n) - H_k] \quad (n \neq 0), \quad H_0 = 0.$$

For all $k, l \geq 0$, one has the distribution identity

$$P_k * \check{P}_l = (\partial_\alpha + \gamma - H_k)(\partial_\beta + \gamma - H_l) \mathcal{N}_{\alpha, \beta} \big|_{\alpha=k, \beta=l}. \quad (6.3)$$

For the derivative-compatible normalization $Q_k = D^k P_0$, the same formula holds with H_k, H_l replaced by zero.

Proof. If $\Re(\alpha + \beta) < -1$, the Fourier series is absolutely convergent. The positive and negative indices separately give (6.2). The same argument applies after parameter differentiation on any smaller open region. The right side has an entire continuation in its orders for $z \neq 1$, and is smooth in h on compact subintervals of $(0, 1)$. To identify its distributional extension, insert the Abel factor $\rho^{|n|}$, $0 < \rho < 1$, and take its limit in distributions. The Hurwitz Fourier formula identifies the same local analytic continuation. The identity therefore extends to all α, β . This argument also specifies terms supported at $h = 0$, which a pointwise formula alone could not recover.

Each parameter derivative in (6.1) multiplies the coefficient by the corresponding logarithm. At (k, l) , this gives $\widehat{P}_k(n) \widehat{P}_l(-n)$, exactly the Fourier coefficient of $P_k * \check{P}_l$. Fourier coefficients determine a distribution on the circle. The formula for Q_k follows from its Fourier coefficient $(2\pi i n)^k [\gamma + \log(2\pi i n)]$. $\square$

The sine phases must be retained for complex parameters. A real-part notation would not preserve holomorphic dependence on α, β . At integer orders $\mathcal{N}_{k, l} = (-1)^l D^{k+l} \Delta$. For $k + l > 0$ this is supported at the endpoint; for $k = l = 0$ its open-interval value is -1 . Its parameter derivatives retain additional, nonconstant interior terms. Substituting an integer before differentiating would lose the desired information.

The unreflected kernel is obtained using one spectral parameter: $\mathcal{M}_\alpha = \mathcal{N}_{\alpha, 0}$. Then

$$P_k * P_l = [\partial_\alpha^2 + (2\gamma - H_k - H_l)\partial_\alpha + (\gamma - H_k)(\gamma - H_l)] \mathcal{M}_\alpha \big|_{\alpha=k+l}. \quad (6.4)$$

The equality is checked on each nonzero Fourier coefficient. It also determines all the contact terms.

6.2 First-Stieltjes identities at polylogarithm order zero

Write $c = \gamma + \log(2\pi)$, $d = \pi/2$, and

$$L_j(h) = \partial_s^j \text{Li}_s(e^{2\pi i h}) \Big|_{s=0}.$$

These derivatives are values of the analytically continued function. They are not defined by directly summing a growing logarithmic Fourier series.

Corollary 6.2 (Two explicit order-derivative reductions). *For every $0 < h < 1$,*

$$\gamma_1(h) + \gamma_1(1-h) = 2\Re L_2(h) - 4c\Re L_1(h) - c^2 + \frac{\pi^2}{12}, \quad (6.5)$$

$$\gamma_1(h) = \Re L_2(h) - 2c\Re L_1(h) + \pi\Im L_1(h) - \frac{c^2}{2} + \frac{\pi^2}{24} - \frac{\pi c}{2} \cot(\pi h). \quad (6.6)$$

Proof. At $k = l = 0$, the operator on the positive-frequency part of (6.2) is

$$(c + di - \partial_s)(c - di - \partial_s) = (c - \partial_s)^2 + d^2.$$

For real h the negative-frequency part is its conjugate. The reflected convolution identity proved above reads on the open interval

$$\gamma_1(h) + \gamma_1(1-h) - \frac{\pi^2}{3} = 2\Re[L_2 - 2cL_1 + (c^2 + \pi^2/4)L_0].$$

Since $\Re L_0 = -1/2$, this is (6.5).

For the unreflected convolution, the operator is $(c + di - \partial_s)^2$. Its value is $2\gamma_1(h) + \zeta(2)$. Finally,

$$L_0(h) = -\frac{1}{2} + \frac{i}{2} \cot(\pi h)$$

gives (6.6) after expanding the real part. Every constant comes from an explicit Gamma coefficient or a contact term. $\square$

These formulas belong to the classical differentiated Hurwitz functional-equation framework. Their role here is an explicit application of the finite-part normalization. They also check the sign of the $\cot(\pi h)$ contribution and the π^2 constants.

At $h = 1/2$, the identity $\text{Li}_s(-1) = (2^{1-s} - 1)\zeta(s)$ turns the corollary into an identity among $\zeta^{[j]}(0)$, γ_1 , γ , and $\log 2$. On the Stieltjes side,

$$\gamma_1(1/2) = \gamma_1 - 2\gamma \log 2 - \log^2 2,$$

obtained directly by expanding $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$. Thus the convergent digamma convolution from the preceding part, at $a = 1/2$, has the explicit value

$$2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 + \frac{\pi^2}{6}. \quad (6.7)$$

This evaluates the fully subtracted integral; the corresponding unsubtracted product integral diverges.

6.3 Rational shifts and a finite coordinate algorithm

The pending report asks for rational-shift coordinate formulas [2]. The conversion follows from a finite Fourier transform already present in the manuscript. The distributional extension now makes it applicable to mixed polygamma kernels at singular spectral orders.

For $q \geq 2$, $1 \leq p < q$, and $z = e^{2\pi ip/q}$,

$$\text{Li}_s(z) = q^{-s} \sum_{a=1}^q z^a \zeta(s, a/q). \quad (6.8)$$

At any regular spectral point s_0 ,

$$\text{Li}_{s_0}^{[j]}(z) = q^{-s_0} \sum_{a=1}^q z^a \sum_{v=0}^j \binom{j}{v} (-\log q)^{j-v} \zeta^{[v]}(s_0, a/q). \quad (6.9)$$

At $s = 1$ the pole cancels because $\sum_{a=1}^q z^a = 0$; one must expand the finite combined expression before specializing. Equations (6.2)–(6.9) are a finite algorithm for every rational-shift polygamma convolution and every fixed spectral derivative.

A multiplicative-character transform then converts these Hurwitz coordinates to primitive Dirichlet- L coordinates. Conductor logarithms are generated by differentiating the explicit factors. The manuscript's conductor-lifted trace theorem and pole-corrected jets already prove that arithmetic conversion [1]. We use them as an available normal form and do not claim independence of the resulting evaluated constants.

7 Source corrections and the scope of the conclusions

The following audit concerns the pinned manuscript, rather than earlier versions already corrected there. It distinguishes a branch error, an unsupported description of a special value, and an editorial clarification about coefficient fields. The accompanying patch changes only the first two items; it does not alter the source files in this research package.

7.1 A real branch correction for the arctanh integral

In `chapters/03-algebraic.tex`, equation `cleo:eq:lintanh` occurs in the discussion of real parameters of absolute value less than one. It writes

$$\int_0^{\pi/2} \text{arctanh}(\lambda \sin x) dx = \theta \log \tan \frac{\theta}{2} + 2 \text{Cl}_2(\theta) - \frac{1}{2} \text{Cl}_2(2\theta), \quad \theta = \arcsin \lambda.$$

The displayed logarithm is correct for positive λ . For negative λ , its argument is negative, and the principal complex logarithm adds an imaginary term that cannot occur in the real integral.

Proposition 7.1 (The real-parameter formula). *For $-1 \leq \lambda \leq 1$, put $\theta = \arcsin \lambda$. Then*

$$\int_0^{\pi/2} \text{arctanh}(\lambda \sin x) dx = \theta \log \left| \tan \frac{\theta}{2} \right| + 2 \text{Cl}_2(\theta) - \frac{1}{2} \text{Cl}_2(2\theta). \quad (7.1)$$

At $\lambda = 0$ the right side is its continuous value zero. At $\lambda = \pm 1$ the integrals are convergent improper integrals.

Proof. Let the integral be $F(\lambda)$. For $|\lambda| < 1$, differentiation under the integral is valid locally uniformly in λ . The substitution $t = \cos x$ gives

$$F'(\lambda) = \int_0^1 \frac{dt}{1 - \lambda^2 + \lambda^2 t^2} = \frac{\arcsin \lambda}{\lambda \sqrt{1 - \lambda^2}},$$

where the quotient has value one at zero. Therefore $dF(\sin \theta)/d\theta = \theta / \sin \theta$.

The classical Fourier definition of the Clausen function gives $\text{Cl}_2'(u) = -\log |2 \sin(u/2)|$ away from multiples of 2π . On either interval $(-\pi/2, 0)$ or $(0, \pi/2)$, differentiating the right side of (7.1) gives $\theta / \sin \theta$: the three logarithmic terms cancel by $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$. Both sides tend to zero at the origin, proving the identity on $(-1, 1)$. Finally $\text{arctanh}(\sin x)$ has only an integrable logarithmic singularity at $x = \pi/2$; dominated convergence and oddness give the endpoint cases. $\square$

The failure at a specific admissible negative parameter is exact. At $\lambda = -1/2$, $\theta = -\pi/6$, the printed principal-logarithm right side differs from the integral by

$$i\pi\theta = -\frac{i\pi^2}{6}.$$

The real integral is approximately $-0.5317299165586139653837501727942413753$. Thus the correction is a branch issue, not a numerical uncertainty. The smallest repair is to insert the absolute value in the logarithm and specify the continuous value at zero. Restricting the printed identity to $0 < \lambda < 1$ would also make its original logarithm valid.

7.2 Remove a remaining unsupported non-elementarity label

The same chapter, at line 194 of the pinned source, calls

$$\int_0^{\pi/2} \arctan^2(\sin x) dx = \pi\chi_2(3 - 2\sqrt{2})$$

the “simplest non-elementary diagonal.” The exact evaluation is retained. The manuscript gives neither a specified comparison algebra nor a proof that this constant lies outside one. Its later discussion correctly distinguishes exact formulas from unproved minimality and non-elementarity. The proposed patch replaces the unsupported adjective by an explicit statement that no elementary reduction or nonreduction theorem for this value is established there. This is a correction of the claim’s status; it is not a proof of a new reduction.

7.3 Coefficient fields depend on the coordinates

The paragraph following `integral:res:parity` in `chapters/07-integration.tex` says that the reduction coefficients live in $\mathbb{Q}(\sqrt{q})$. That description fits the immediately displayed real quadratic character modulo five. It should not be read as a field assertion for arbitrary rational grids or arbitrary characters. Chapter 2 already correctly uses the cyclotomic DFT field $\mathbb{Q}(\zeta_q)$; passing to multiplicative characters can also introduce character values and Gauss sums.

For example, the primitive character modulo five specified by $\chi(2) = i$ takes the values $1, i, -i, -1$ on $1, 2, 3, 4$. The character-coordinate coefficients therefore include i , which is not in the

real field $\mathbb{Q}(\sqrt{5})$. Even a real sine-coordinate presentation need not stay in that quadratic field:

$$\sin^2 \frac{2\pi}{5} = \frac{5 + \sqrt{5}}{8}, \quad N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}} \left(\sin^2 \frac{2\pi}{5} \right) = \frac{5}{16}.$$

If $\sin(2\pi/5)$ belonged to $\mathbb{Q}(\sqrt{5})$, the last quantity would be the square of a rational norm, which it is not. The displayed quadratic-character identity itself requires no repair. The appropriate clarification is to state the coefficient field for the chosen additive, character, or real-paired coordinates.

7.4 Prior in-repository continuations must be credited

The pinned repository already contains substantial periodic Stieltjes calculus outside the manuscript chapters, under `docs/reports/stieltjes-correlation-calculus/continuations/` relative to `Analysis/Polylogarithms/` [3, 4, 5]. In particular, `convolution-algebra/article.tex` proves the one-generator Bell algebra in `thm:bell`, the Gamma-quotient multiplication law in `thm:product`, the polygamma contact correction, and the reflected spectral kernel in `thm:reflected`. The file `convolution-calculus/Stieltjes_Convolution_Calculus.tex` already records the symmetric first-Stieltjes order-zero polylogarithm formula in `eq:Li0identity`. The file `shifted-hurwitz-jets/sections.tex` gives the complete normalized primitive tower in `shj:Uformula` and proves its endpoint regularity.

The untwisted periodic results identified above therefore consolidate and independently verify those untwisted foundations. The nonintegral-twist section separately answers the explicit open target in the archived `convolution-algebra` report. The appearance of a question in an earlier intake report does not establish its current open status when a later in-repository continuation has already answered it. The shifted harmonic Laurent and nonintegral-twist results are assessed separately, with their own source and literature qualifications, rather than deriving a novelty claim from the untwisted periodic construction.

An explicitly open target in the same `convolution-algebra/article.tex` is the subsection “Twists, characters, and alternate endpoint normalizations.” It asks for the nonintegral frequency-shift analogue and an all-index half-twist table with its point-supported terms. Section 4 answers this specified target: the convolution unit is δ_0 , the scalar Gamma coefficients remain ordinary zeta values, the basis consists of Lerch jets, and the covariant contact terms are given explicitly by Theorem 4.4. The nonsingular alternating convolution and the modified Stieltjes functions themselves have the antecedents cited there. The resolved repository question concerns their singular extension and normalization.

7.5 What this continuation does not prove

At the pinned snapshot, the mixed sums S_2 and S_4 already have exact proofs. The weight-seven relation `cycloquot:conj:S6` and the newer weight-nine relation `s8new:conj:S8` remain conjectural. A different frozen weight-nine vector was rigorously rejected in `research:prop:S8-rejected`; its rejection must not be confused with the status of the newer candidate. No result of the present continuation promotes either surviving numerical candidate to a theorem.

The distributional identities proved above fix their extensions through meromorphic continuation, or equivalently the stated cutoff finite parts. Agreement with an ordinary function away from the endpoint and zero mean alone do not characterize every higher derivative extension:

derivatives of the endpoint delta distribution have zero mean. The explicit endpoint terms in those theorems are part of the statements and must be preserved when the formulas are integrated or differentiated.

8 Verification, integration, and further research

8.1 What has been proved and what has been checked

The proofs in this article establish the identities. Their main analytic steps are normal convergence after pairing Fourier coefficients with a smooth test function, explicit Taylor subtraction at a singular endpoint, joint meromorphic continuation before taking parameter coefficients, and finite Gamma/Bernoulli coefficient algebra. None of the theorems is inferred from a numerical fit.

The most delicate points were checked independently of their initial derivations. These include the constant Fourier mode, the signs of the covariant Dirac derivatives, the minus sign on the wrapped half-twist interval, the harmonic Mellin remainder at a Gamma zero, the depth-mixing reflection shift $a + u$, and the terminal $\mathcal{A}_{-1} = 1/\Gamma(1 + u)$ term in rational multiplication. The full mathematical review is supplied in `verification/independent_review.md`. This is an independent derivation within the present research workflow, not a claim of external peer review or proof-assistant formalization.

The exact and numerical replays have different roles. Exact symbolic checks verify finite polynomial consequences of the stated generating identities. Numerical checks compare independent analytic representations, with endpoint cancellations removed before evaluation. They are floating-point diagnostics, not rigorous interval enclosures.

Verification component	Recorded coverage
Untwisted coefficient algebra	Six explicit Stieltjes products; reflected phase coefficients through degree eight for both signs; 128 polygamma product cases and 30 repeated-primitive recurrences.
Harmonic exact algebra	167 checks: local kernels, Hurwitz–Bernoulli reductions, residue differentiation, reflection, primitives, multiplication for moduli two and three, and all displayed specializations.
Periodic quadrature	15 checks at 50 decimal digits, including finite-part test-function pairings and ordinary digamma/log-Gamma convolutions.
Harmonic Mellin continuation	Three regular-value checks and four Laurent cases at each of $\varepsilon = 10^{-4}, 10^{-5}, 10^{-6}$, using an independent scalar continued Mellin evaluator at 110 decimal digits.
Half-twist and source repair	Nine checks at 60 decimal digits: the convergent half-twist integral at three shifts, its quarter-Gamma evaluation, four branch-corrected arctanh integrals, and the exact spurious imaginary term.
Twisted spectral normalization	The separate twisted replay tests rational Lerch conversion, the retained zero mode, and the covariant contact coefficients. Its complete parameters and residuals are recorded with the script.

For the harmonic continuation, the scalar integrand is

$$\frac{e^{-(a+1)t}}{1 - e^{-t}} \frac{[-\log(1 - e^{-t})]^r}{r!}.$$

The independent evaluator integrates its convergent local logarithmic series on $(0, 1/2)$, after analytic continuation of the individual Mellin monomials, and performs direct quadrature on $(1/2, \infty)$. It does not evaluate through the $\mathcal{A}_m$ coefficient theorem. Degrees 80 and 100 agreed to better than 1.7×10^{-92} in the recorded tests. After subtracting the asserted principal and constant terms, the remainder was $O(\varepsilon)$ in every tested case. That agreement diagnoses the coefficient extraction; it does not certify a truncation bound for arbitrary inputs.

The extra half-twist quadrature uses only digamma and polygamma functions on the integral side, and Hurwitz Stieltjes constants or $\log \Gamma(1/4)$ on the comparison side. A Taylor quotient removes the cancellation at $x = 0$. Its largest recorded scaled residual, including the branch-audit checks, is below 4×10^{-61} . Here a scaled residual means

$$\frac{|\text{computed left side} - \text{computed right side}|}{1 + |\text{computed right side}|}.$$

A tiny residual is useful implementation evidence, not an equality proof.

8.2 Suggested repository integration

The package is designed for archival intake first and mathematical integration second. The supplied source correction patch applies against the pinned chapter bytes. The main article has modular sources and an equivalent standalone source, plus complete replay scripts and recorded results.

The integration should preserve the following distinctions:

1. Cite the existing untwisted continuations instead of replacing their provenance with the present rederivation.
2. Add the nonintegral-twist section as a response to the archived half-twist question, retaining its cutoff definition, covariant derivative, unit δ_0 , and reflection modulation.
3. Add the shifted harmonic Laurent theorem together with its precise outer-shift convention. Moving the inner harmonic denominators as well defines a different family.
4. Keep each finite-part identity attached to its domain and normalization. An isolated ordinary formula does not determine its endpoint distribution.
5. Apply the two narrowly scoped source corrections separately from accepting the new research statements.

The machine-readable claim ledger records proved additions, consolidated antecedents, corrections, and unresolved targets. The proposed patch does not change the status of S_6 or S_8 .

8.3 Further research questions

The following questions concern exact identities and reductions. They are proposals for further work, not claims supported merely by high-precision numerical evidence.

Question 8.1 (Products with genuinely different twists). For real nonintegral θ, η , with $\theta - \eta \notin \mathbb{Z}$, determine a useful finite or convergently evaluable normal form for

$$\widehat{\mathcal{K}}_{\alpha, \beta}^{\theta, \eta}(n) = (2\pi i(n + \theta))^\alpha (2\pi i(n + \eta))^\beta.$$

Which parameter configurations permit reduction to finitely many single-twist Lerch jets, and when is a genuinely two-shift special function necessary?

The equal-twist case is now solved by the Gamma quotient. The half-twist involution reverses frequencies within one twist; it does not solve the unequal-twist problem. A successful extension should retain its Laurent endpoint terms and rational-character coordinates, rather than only its nonsingular Fourier expression.

Question 8.2 (Regular Laurent coefficients beyond the finite part). For $j \geq 1$, determine exact representations and reduction rules for

$$[\varepsilon^j]E_r(-m + \varepsilon; a).$$

At rational a , when can these coefficients be expressed using ordinary Hurwitz or Dirichlet- L derivatives, and when do new height-one or colored multiple-zeta derivatives remain?

The Gamma zero eliminates the analytic Mellin remainder through the constant coefficient, but not beyond it. Thus the polynomial theorem proved here gives the full singular and finite data without settling the next regular coefficient. The independent scalar Mellin representation supplies a rigorous starting point for such a study.

Question 8.3 (Lifting the reflection and multiplication laws). Find exact formulas for the analytic defects obtained when (5.48) and (5.53) are lifted from Laurent coefficient polynomials to the full meromorphic family $E_r(s; a)$. Can the defects be organized as convergent iterated integrals or colored polylogarithms with a natural depth filtration?

The finite polynomial identities are exact, not asymptotic. Nevertheless their proof uses only the local Mellin kernel. A global lift must also control the rest of the integral. Keeping that local/global separation explicit should prevent a false unmodified Hurwitz multiplication rule for the harmonic family.

Question 8.4 (Rational-twist arithmetic at singular orders). Combine the all-index twisted table with the manuscript's conductor-sensitive character conversion. Determine economical primitive-character formulas for the resulting rational-shift convolutions, including every conductor logarithm and endpoint term. Which combinations descend to a proper sub-conductor?

A finite coordinate algorithm is already available; the remaining problem is its exact arithmetic simplification. Reduction to a chosen spanning set is a meaningful target even when independence of the evaluated constants is unknown.

Question 8.5 (Translated cubic and higher log-Gamma products). With $g(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$, obtain exact reductions for the two-parameter ordinary product

$$C_3(a, b) = \int_0^1 g(x)g(\{x + a\})g(\{x + b\}) dx, \quad 0 < a < b < 1,$$

and for higher moments beyond the known unshifted cubic formula. At rational a, b , which colored multiple-zeta derivatives remain after exact character and symmetry reductions?

The unshifted cubic moment is already evaluated in the pinned chapters 07-loggamma-moments.tex and 07-tornheim-evaluation.tex, and must not be proposed as an unsolved problem [1].

The target here retains two distinct translation parameters. These are pointwise products, distinct from convolution powers: Fourier multiplication introduces several independent indices. A concrete starting family is the meromorphic continuation of

$$\mathcal{T}(s, t, u; z, w) = \sum_{m, n \geq 1} \frac{z^m w^n}{m^s n^t (m + n)^u},$$

and its order derivatives at the integer triples arising from Kummer's Fourier coefficients, with phases fixed by a, b . The research target is an explicit reduction theorem, not a numerical assertion that a selected constant basis is complete or independent.

Question 8.6 (Collision regularizations across different products). When translated singularities collide, compare the finite part obtained from the meromorphic convolution family with finite parts obtained by coincident-point cutoff procedures. Determine the exact local counterterm relating these prescriptions, and which associativity or differentiation identities each preserves.

The distribution convolution itself is defined on the compact circle. Evaluating it at its singular point is a different operation and needs a separate prescription. The zero-mode degeneration theorem controls a twist resonance, not every possible collision regularization of pointwise products.

Question 8.7 (The remaining Gaussian candidates). Can the level-four iterated-integral and conductor-jet reductions in the manuscript be completed into an exact certificate for `cycloquot:conj:S6` or `s8new:conj:S8`? If a conjectured coefficient vector is false, can it be rejected with a certified error enclosure and replaced by a precisely stated new target?

The source already distinguishes these candidates from the proved S_2, S_4 identities and the rejected earlier S_8 vector. The present twist and harmonic identities should be used as additional transformations, without suggesting that their existence proves either outstanding relation.

A practical formalization programme. A useful accompanying project is to formalize the finite coefficient algebra separately from the analytic continuation lemmas. The exact Bell, Bernoulli, reflection, multiplication, and residue recurrences give compact symbolic certificates at any requested depth. The analytic layer should then discharge the local-uniform convergence, endpoint subtraction, and uniqueness hypotheses once, instead of replacing them by growing collections of numerical examples.

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